

Linear Algebra 101

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This lecture note is a work in progress, and will be updated from time to time.

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1 Motivation

Before introducing the abstract mathematics of vector spaces, it is worth asking *why* quantum mechanics needs this language at all. Two experiments — one genuinely quantum (Stern–Gerlach) and one with a familiar classical description (the polarization of light) — motivate the vector-space description of physical states, and it is this analogy, more than any axiom, that makes the formalism developed in the rest of these notes feel inevitable rather than arbitrary. (As section 1.4 makes explicit, the second example is only classical in a particular, macroscopic limit; at the level of individual photons it is every bit as quantum mechanical as the first.)

1.1 The Stern–Gerlach Experiment (1922)

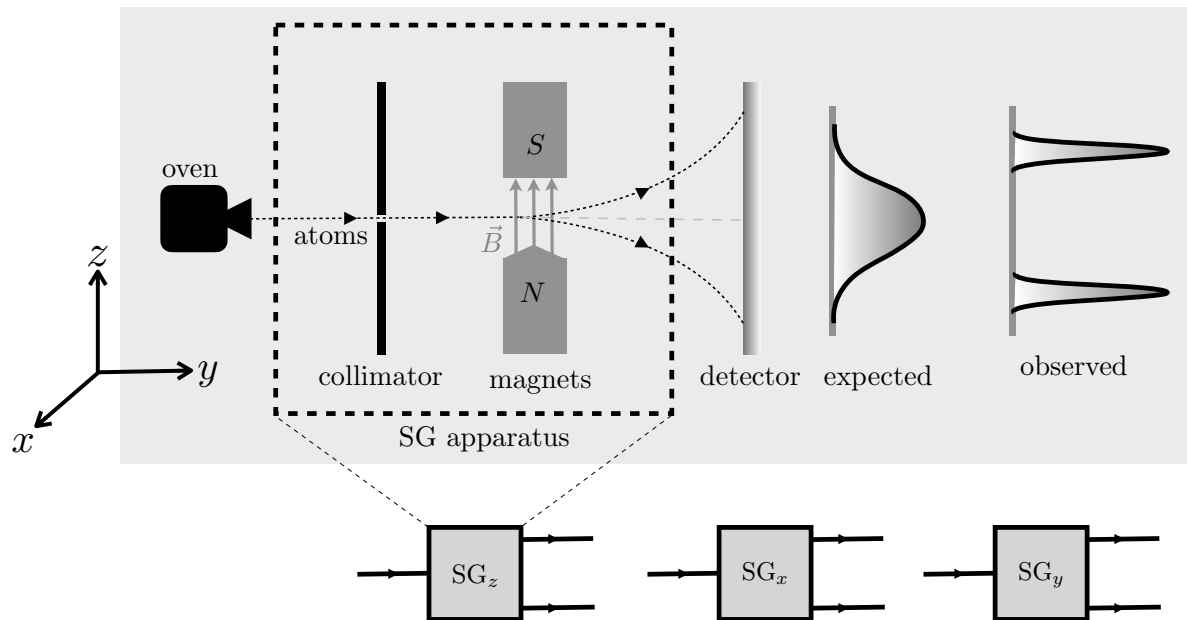


Figure 1: The Stern–Gerlach apparatus. A collimated beam of silver atoms passes through an inhomogeneous magnetic field $\mathbf{B} \approx B_z \hat{z}$ produced by the pole pieces N, S . Classically, a continuous spread of orientations of the magnetic moment $\boldsymbol{\mu}$ should produce the smooth “expected” distribution sketched on the screen; what is actually “observed” is only two discrete spots. The three boxes at the bottom introduce the shorthand SG_z, SG_x, SG_y for an SG analyzer oriented along a given axis, notation used throughout the rest of this section.

A beam of silver atoms, each carrying a magnetic moment $\boldsymbol{\mu} = (\mu_x, \mu_y, \mu_z)$ of the same magnitude, is collimated and passed through an *inhomogeneous* magnetic field $\mathbf{B} \approx B_z \hat{z}$ (with $B_x, B_y \rightarrow 0$). The potential energy and the resulting force on a magnetic dipole are

$$U = -\boldsymbol{\mu} \cdot \mathbf{B} = -\mu_z B_z, \quad \mathbf{F} = F_z \hat{z} = \mu_z \frac{\partial B_z}{\partial z} \hat{z}, \quad (1)$$

where $\partial B_z / \partial z$ is assumed constant across the beam.

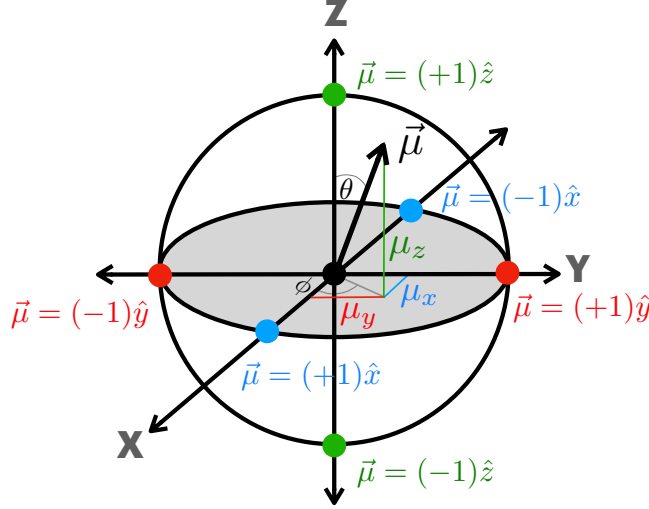


Figure 2: Classically, the magnetic moment $\boldsymbol{\mu}$ of a silver atom can point in any direction on a sphere, parametrized by the polar and azimuthal angles θ, ϕ , so that its components μ_x, μ_y, μ_z can each take any value continuously between -1 and $+1$ (in units of $|\boldsymbol{\mu}|$) — in contrast to the two discrete values actually observed along any given axis.

Classically the orientation of $\boldsymbol{\mu}$ is random (fig. 2), so μ_z can take *any* value in $-1 \leq \mu_z \leq 1$ (with $0 \leq |\boldsymbol{\mu}| \leq 1$), and one would expect a continuous smear of silver deposits on the screen between the two extremes $\mu_z = +1$ and $\mu_z = -1$, as sketched in the “expected” panel of fig. 1.

What Stern and Gerlach observed instead was two *discrete* spots, corresponding only to $\mu_z = +1$ and $\mu_z = -1$ (denoted z_+ and z_-). This is the first hint that a component of angular momentum along a given axis is quantized, and it is completely inexplicable within classical mechanics: whatever variable μ_z actually is, it does not behave like an ordinary continuous classical quantity.

1.2 Sequential Stern–Gerlach Measurements

The puzzle deepens once several Stern–Gerlach (SG) analyzers, in the notation of fig. 1, are placed one after another.

In the first arrangement, a beam pre-selected to be pure z_+ by an initial $\text{SG}_z^{(1)}$ analyzer is passed through a second analyzer, $\text{SG}_z^{(2)}$, again oriented along z : as classical intuition demands, once μ_z is fixed it stays fixed, and only z_+ emerges. In the second arrangement, the same pure z_+ beam is instead sent into $\text{SG}_x^{(2)}$, oriented along x , and — despite every atom entering with $\mu_z = +1$ — the x -measurement still yields *both* x_+ and x_- . In the third arrangement, blocking the z_- beam after the first analyzer should (classically) guarantee $\mu_z = +1$ forever; yet after an intervening x -measurement, a z_- component *reappears* once the beam is passed through a third analyzer $\text{SG}_z^{(3)}$. Selecting x_+ has evidently destroyed the earlier information about μ_z , and no classical theory — in which a particle simply carries a fixed, if unknown, value of $\boldsymbol{\mu}$ at all times — can explain the outcome of such sequential measurements.

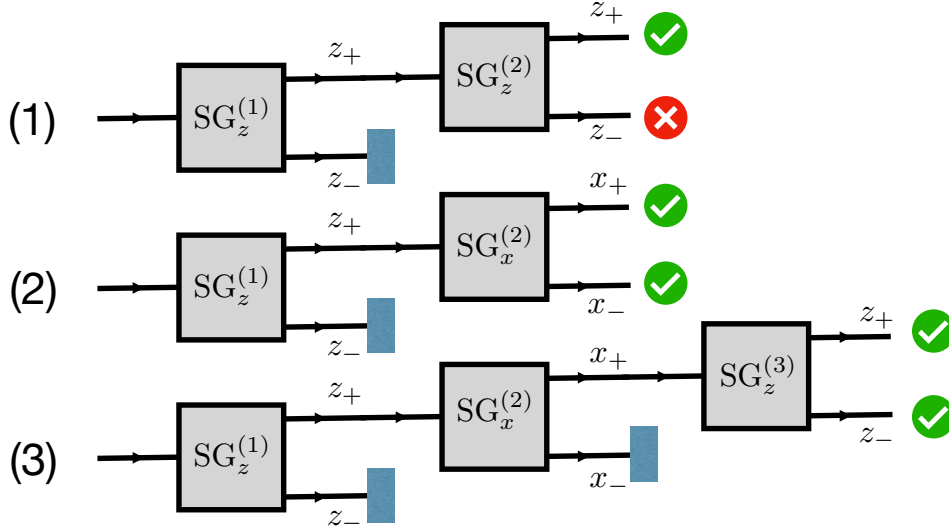


Figure 3: Sequential Stern–Gerlach measurements. (1) A beam pre-selected to be pure z_+ by a first SG_z analyzer is re-measured by a second SG_z analyzer. (2) The same pure z_+ beam is instead measured along x . (3) The x_+ component selected in (2) is re-measured along z .

1.3 Classical Analogy: Polarization of Light

A completely classical phenomenon (not happy? See section 1.4) — the polarization of light — exhibits strikingly similar behaviour, and turns out to be the right classical analogy to keep in mind while building the quantum formalism.

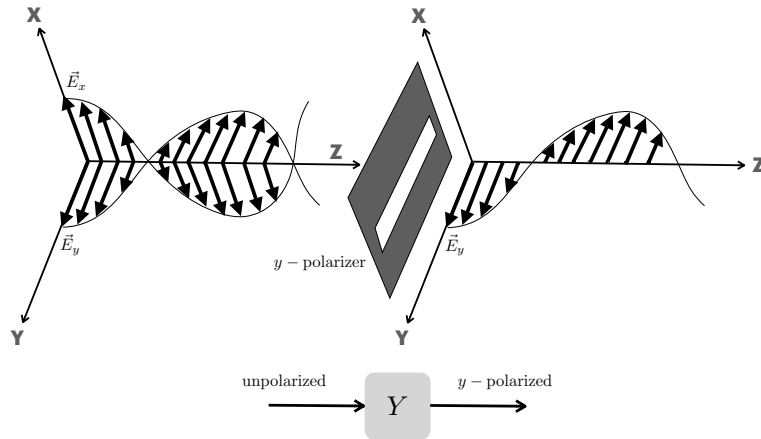


Figure 4: Unpolarized light — an incoherent superposition of oscillating electric-field components $\mathbf{E}_x, \mathbf{E}_y$ — passed through an ideal y -polarizer Y emerges purely y -polarized: only the $\mathbf{E}_y$ component survives.

Now send the linearly polarized light through a sequence of ideal polarizers, exactly as we did with the SG analyzers:

Panel (c) of fig. 5 is the classical counterpart of the $SG_z \rightarrow SG_x \rightarrow SG_z$ puzzle of fig. 3: inserting an intermediate polarizer X' tilted at $\theta = \pi/4$ between crossed polarizers X and Y lets light through, even though X and Y alone pass nothing, as panel (b) shows.

This is understood classically because the electric field is a *vector*; resolving it along the

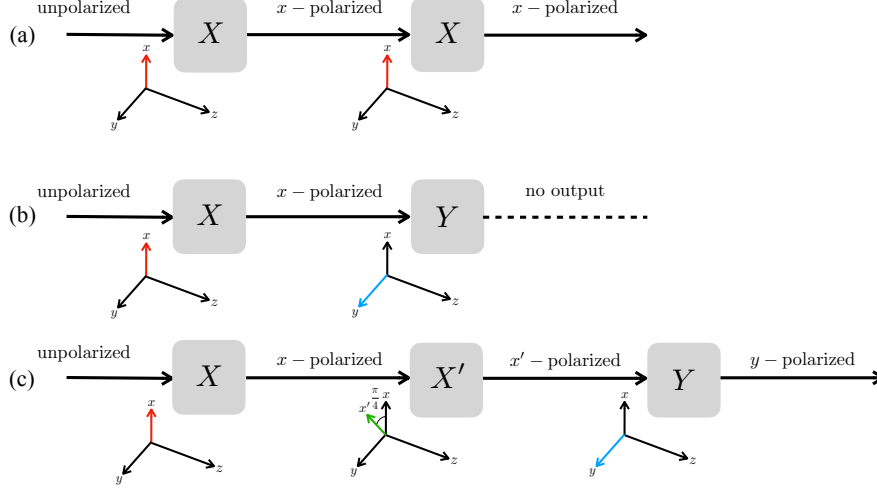


Figure 5: Sequences of ideal polarizers, in exact analogy with fig. 3. (a) A pure x -polarized beam passes unchanged through a second x -polarizer. (b) The same beam is completely blocked by a crossed (y) polarizer. (c) Inserting an intermediate polarizer X' , tilted by $\theta = \pi/4$ relative to x , between the crossed X and Y polarizers allows some light through after all.

rotated axes $\hat{x}', \hat{y}'$ of fig. 5(c) (with $\hat{x}' = (\hat{x} + \hat{y})/\sqrt{2}$, $\hat{y}' = (\hat{y} - \hat{x})/\sqrt{2}$) gives

$$\mathbf{E}_{x'} = E_0 \cos(kz - \omega t) \hat{x}' = E_0 \cos(kz - \omega t) \left(\frac{\hat{x} + \hat{y}}{\sqrt{2}} \right), \quad (2)$$

$$\mathbf{E}_{y'} = E_0 \cos(kz - \omega t) \hat{y}' = E_0 \cos(kz - \omega t) \left(\frac{\hat{y} - \hat{x}}{\sqrt{2}} \right), \quad (3)$$

so that, inverting,

$$\mathbf{E}_x = E_0 \cos(kz - \omega t) \left(\frac{\hat{x}' - \hat{y}'}{\sqrt{2}} \right), \quad (4)$$

$$\mathbf{E}_y = E_0 \cos(kz - \omega t) \left(\frac{\hat{x}' + \hat{y}'}{\sqrt{2}} \right) \quad (\text{has both } x' \text{ and } y' \text{ components}). \quad (5)$$

Light polarized along x (i.e. $\hat{x}$ -polarized), when re-expressed in the $x'y'$ basis, has *both* an x' and a y' component; the X' polarizer projects out the x' piece, which itself has a nonzero y -component, so some light survives all the way to Y . Thinking of the polarization state as a *vector*, and of each polarizer as a device that projects that vector onto a one-dimensional subspace, resolves what looked like a paradox.

The moral to draw from this classical analogy is exactly the one we will need: polarization states of light behave like vectors in a two-dimensional space, and passing light through a polarizer is a projection onto a one-dimensional subspace of that space. The Stern–Gerlach outcomes ($z_{\pm}$, $x_{\pm}$, and so on) turn out to behave the same way, except that the underlying vector space is over the complex numbers, and the “lengths” that matter are probabilities (amplitudes squared) rather than field intensities. This is exactly the structure the rest of these notes make precise: a complex linear vector space equipped with an inner product, i.e. a **Hilbert space**.

1.4 How Classical Is This, Really?

It is worth being explicit about exactly where classical physics stops doing the work in the analogy above, since the phrase “classical analogy” is easy to over-read. Everything in figs. 4 and 5 and the electric-field algebra following them describes light as a classical electromagnetic

wave: $\mathbf{E}$ is an ordinary real three-vector, a polarizer acts on the continuous field amplitude, and the fraction of intensity transmitted is fixed by Maxwell's equations alone. In this regime — essentially any everyday beam of light, made up of an enormous number of photons — polarization is exhaustively explained without invoking quantum mechanics at all, which is exactly why it could be used above as a purely classical stepping stone toward the vector-space language.

That is not, however, the whole story of polarization, only the part needed for the analogy. If the same polarizer experiments are repeated one photon at a time, each individual photon is found to pass through a polarizer whole or not at all: there is no such thing as a fraction of a photon being transmitted. The *probability* that a given photon is transmitted through, say, the X' polarizer of fig. 5(c) is given by exactly the projection formula derived above for the classical field amplitude, now reinterpreted as a quantum-mechanical probability rather than a fraction of field intensity — this is the single-photon (quantum) form of Malus's law. In this regime a photon's polarization state is itself a genuine two-level quantum system, living in a two-dimensional complex Hilbert space in exactly the sense to be developed below, with $|x\rangle$ and $|y\rangle$ (or, equally validly, $|x'\rangle$ and $|y'\rangle$) as an orthonormal basis of that space. Photon polarization is, in fact, one of the two standard laboratory settings — alongside spin- $\frac{1}{2}$, the actual Stern–Gerlach observable of section 4.13 — in which entanglement and violations of Bell's inequalities have been experimentally demonstrated, phenomena with no classical explanation whatsoever.

The classical wave picture used above, in short, correctly reproduces the statistics of a beam containing many photons, but it is the quantum, single-photon description that is fundamental; the two agree on every observable prediction only because probabilities (amplitudes squared) at the single-photon level average, over a large number of photons, to the classical intensities. Polarization was chosen here as the classical analogy precisely because this same mathematical structure — projection of a state vector onto a one-dimensional subspace — already appears at the classical level, making it the natural bridge to the fully quantum vector-space formalism that both it and Stern–Gerlach ultimately require. We now turn to building that structure from first principles.

2 Vectors and Linear Vector Spaces

2.1 Fields

A *field* $\mathcal{F}$ is a set of numbers, equipped with addition and multiplication, satisfying the following properties in full. Elements of a field are commonly referred to as *scalars*.

(1) **Addition:** For $a, b \in \mathcal{F}$ there exists a sum $a + b \in \mathcal{F}$ such that

$$(a) \quad a + b = b + a \quad \text{(commutative)}$$

$$(b) \quad a + (b + c) = (a + b) + c \quad \text{(associative)}$$

$$(c) \quad \text{there exists an additive identity } 0 \in \mathcal{F} \text{ such that } a + 0 = a \text{ for all } a \in \mathcal{F},$$

$$(d) \quad \text{every } a \in \mathcal{F} \text{ has an additive inverse } -a \in \mathcal{F} \text{ such that } a + (-a) = 0.$$

(2) **Multiplication:** For $a, b \in \mathcal{F}$ there exists a product $ab \in \mathcal{F}$ such that

$$(a) \quad ab = ba \quad \text{(commutative)}$$

$$(b) \quad a(bc) = (ab)c \quad \text{(associative)}$$

$$(c) \quad \text{there exists a multiplicative identity } 1 \in \mathcal{F}, \text{ with } 1 \neq 0, \text{ such that } 1 \cdot a = a \text{ for all } a \in \mathcal{F},$$

$$(d) \quad \text{every } a \in \mathcal{F} \text{ other than } 0 \text{ has a multiplicative inverse } a^{-1} \in \mathcal{F} \text{ such that } aa^{-1} = 1.$$

- (3) **Distributivity:** multiplication distributes over addition, $a(b+c) = ab+ac$, for all $a, b, c \in \mathcal{F}$.

This list is exhaustive: any set with two operations satisfying (1)–(3) is, by definition, a field, and conversely every property a field enjoys follows from these axioms. Of the examples below, it is $\mathbb{C}$ and $\mathbb{R}$ that will occupy us almost exclusively.

Symbol	Field
$\mathbb{C}$	complex numbers
$\mathbb{R}$	real numbers
$\mathbb{Q}$	rational numbers

It is worth pausing on a near-miss: the integers $\mathbb{Z}$ satisfy every axiom above *except* (2d), since, for instance, $2 \in \mathbb{Z}$ has no integer a with $2a = 1$. The multiplicative-inverse axiom is therefore doing real work, not merely padding the list — its absence is exactly what separates a field from the weaker structure $\mathbb{Z}$ possesses (a *ring*, in the standard terminology, meaning (1) and (3) hold but (2d) is dropped). Indeed $\mathbb{Q}$ is, in a precise sense, the smallest field built from $\mathbb{Z}$: it is obtained by formally adjoining a multiplicative inverse $1/n$ for every nonzero $n \in \mathbb{Z}$. This is not merely a pedantic aside: it is the reason the vector spaces of this chapter are built over $\mathbb{C}$ or $\mathbb{R}$ and never over $\mathbb{Z}$, since scalar multiplication by $1/n$ (needed, for instance, to normalize a vector) requires an honest multiplicative inverse.

2.2 Bra–Ket Notation

Dirac’s notation denotes a vector by a *ket*, $|a\rangle$, and its dual vector by a *bra*, $\langle a|$. We will see shortly that the bra $\langle a|$ is best thought of not as a vector in its own right but as a linear functional that eats a ket and returns a scalar; for now it is enough to treat $|a\rangle$ and $\langle a|$ as two different notations for the same physical state, related by complex conjugation once a basis is chosen.

2.3 Linear Vector Space

A *linear vector space* (LVS) is the space in which vectors live; we denote it by α . It satisfies the following properties.

- (1) **Vector addition:** For $|a_1\rangle, |a_2\rangle \in \alpha$ there exists a vector $|a\rangle = |a_1\rangle + |a_2\rangle \in \alpha$ such that
- (a) $|a_1\rangle + |a_2\rangle = |a_2\rangle + |a_1\rangle$ (commutative)
 - (b) $|a_1\rangle + (|a_2\rangle + |a_3\rangle) = (|a_1\rangle + |a_2\rangle) + |a_3\rangle$ (associative)
 - (c) There exists a null vector $|a_0\rangle \in \alpha$ such that $|a\rangle + |a_0\rangle = |a\rangle$ for all $|a\rangle \in \alpha$.
 - (d) There exists an additive inverse $|a'\rangle = -|a\rangle$ of $|a\rangle$, for every $|a\rangle \in \alpha$, such that $|a\rangle + |a'\rangle = |a_0\rangle$.
- (2) **Scalar multiplication:** For any $|a\rangle \in \alpha$ and any scalar c (real or complex), the vector $c|a\rangle$, obtained by scalar multiplication of c with $|a\rangle$, also lives in α , such that
- (a) $c_1(c_2|a\rangle) = (c_1c_2)|a\rangle$
 - (b) $1 \cdot |a\rangle = |a\rangle$, i.e. 1 is the multiplicative identity
 - (c) $(c_1 + c_2)|a\rangle = c_1|a\rangle + c_2|a\rangle$
 - (d) $c(|a_1\rangle + |a_2\rangle) = c|a_1\rangle + c|a_2\rangle$ (distributive)

The set of bras $\langle a|$ also forms an LVS in its own right — the *dual* of α . Four examples will recur throughout the course and are worth keeping in mind side by side, since much of what follows is designed to treat them all on the same footing.

Ordinary three-dimensional physical space is the most familiar instance,

$$|a\rangle = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix}, \quad \langle a| = (a_x \quad a_y \quad a_z), \quad a_x, a_y, a_z \in \mathbb{R}.$$

More generally, the set of all $n \times 1$ column matrices over $\mathbb{C}$ forms an LVS,

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{pmatrix}, \quad \langle a| = (a_1^* \quad a_2^* \quad a_3^* \quad \cdots \quad a_n^*), \quad a_i \in \mathbb{C} \quad \forall i,$$

and this is the space we will use whenever we need finitely many complex amplitudes, as for a spin degree of freedom. So, too, does the set of all $n \times n$ square matrices over $\mathbb{C}$,

$$|a\rangle = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}, \quad a_{ij} \in \mathbb{C} \quad \forall 1 \leq i, j \leq n,$$

with vector addition being ordinary matrix addition.

The fourth example takes us to infinite dimensions and is the one that will eventually house physical wavefunctions: the set of square-integrable functions on an interval $[a, b]$, meaning those $f(x)$ for which

$$\int_a^b |f(x)|^2 dx < \infty.$$

That this set closes under addition, and is therefore genuinely a linear vector space, is not entirely obvious and is worth checking explicitly. For two square-integrable functions $f(x)$ and $g(x)$,

$$|f + g|^2 + |f - g|^2 = 2|f|^2 + 2|g|^2 \implies |f + g|^2 \leq 2|f|^2 + 2|g|^2,$$

and integrating both sides gives

$$\int_a^b |f(x) + g(x)|^2 dx \leq 2 \left[\int_a^b |f(x)|^2 dx + \int_a^b |g(x)|^2 dx \right] < \infty.$$

Hence $f(x) + g(x)$ is also square integrable, so square-integrable functions on $[a, b]$ do indeed close under addition and form an LVS.

2.4 Spanning Sets, Linear Independence, and Basis

Let $B = \{|b_1\rangle, |b_2\rangle, \dots, |b_n\rangle\}$ be a set of vectors in α . B is a *spanning set* of α if every $|a\rangle \in \alpha$ can be written as

$$|a\rangle = \sum_{i=1}^n c_i |b_i\rangle, \quad c_i \in \mathbb{C} \quad \forall i.$$

The vectors of B are *linearly independent* (LI) if the only solution for $\{c_i\}$ satisfying $\sum_i c_i |b_i\rangle = 0$ is $c_i = 0$ for every i . When B both spans α and is linearly independent, it is called a *basis* of α ; equivalently, a basis is a maximal set of linearly independent vectors in α . The number of

elements in a basis is the *dimension* of α , and one can show it is the same for every choice of basis, even though the basis itself is far from unique, as we see below.

In ordinary three-dimensional space, the vectors $|a_1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $|a_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $|a_3\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ form the familiar basis in which $|a\rangle = a_x |a_1\rangle + a_y |a_2\rangle + a_z |a_3\rangle$ for $a_x, a_y, a_z \in \mathbb{R}$; the analogous standard basis for the $n \times 1$ column matrices over $\mathbb{C}$ consists of $|a_1\rangle = (1, 0, \dots, 0)^T$, $|a_2\rangle = (0, 1, 0, \dots, 0)^T, \dots, |a_n\rangle = (0, \dots, 0, 1)^T$, so that any $|a\rangle = \sum_{i=1}^n a_i |a_i\rangle$ with $a_i \in \mathbb{C}$.

It is worth pausing on the fact — easy to state, easy to underestimate — that a basis of an LVS is never unique. Consider again the space of $n \times 1$ column matrices over $\mathbb{C}$ with the standard basis $B = \{|a_1\rangle, \dots, |a_n\rangle\}$, so that an arbitrary vector is

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \sum_{i=1}^n a_i |a_i\rangle. \quad (1)$$

Now consider instead the set

$$B' = \left\{ |a'_1\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, |a'_2\rangle = \begin{pmatrix} 1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, |a'_n\rangle = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \right\},$$

i.e. $|a'_k\rangle$ has 1's in the first k entries and 0's thereafter. Expanding the same vector in B' ,

$$|a\rangle = \sum_{i=1}^n a'_i |a'_i\rangle = \begin{pmatrix} a'_1 + a'_2 + \dots + a'_n \\ a'_2 + a'_3 + \dots + a'_n \\ \vdots \\ a'_n \end{pmatrix}, \quad (2)$$

and demanding that the vector $|a\rangle$ in (1) and (2) be one and the same gives

$$a'_1 = a_1 - a_2, \quad a'_2 = a_2 - a_3, \quad \dots, \quad a'_{n-1} = a_{n-1} - a_n, \quad a'_n = a_n.$$

Since every $|a\rangle$ can equally well be expanded in B' , the set B' is also a basis of the same space: the basis of an LVS is therefore not unique, and we are free to choose whichever basis is most convenient for the problem at hand.

2.5 Scalar (Inner) Product

To each ordered pair of vectors $|a_1\rangle, |a_2\rangle \in \alpha$ we may associate a complex number $c = \langle a_1 | a_2 \rangle$, called their *inner* or *scalar* product, provided the association satisfies

- (a) $\langle a_1 | a_2 \rangle = \langle a_2 | a_1 \rangle^*$,
- (b) $\langle a_1 | (c_2 |a_2\rangle + c_3 |a_3\rangle) = c_2 \langle a_1 | a_2 \rangle + c_3 \langle a_1 | a_3 \rangle$ for $c_2, c_3 \in \mathbb{C}$,
- (c) $\langle a | a \rangle \geq 0 \quad \forall |a\rangle \in \alpha$, with $\langle a | a \rangle = 0$ iff $|a\rangle = 0$.

An LVS on which an inner product is defined is called an *inner-product space*. It is exactly the property (a) that justifies thinking of the bra $\langle a |$ as living in a different (dual) space from the ket $|a\rangle$, since $\langle a_1 | a_2 \rangle$ and $\langle a_2 | a_1 \rangle$ are related by complex conjugation rather than being equal outright.

Returning to our four running examples, in ordinary three-dimensional space (with $a_{1,i}, a_{2,i} \in \mathbb{R}$) the inner product is the familiar dot product,

$$\langle a_1 | a_2 \rangle = \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \end{pmatrix} \begin{pmatrix} a_{2,1} \\ a_{2,2} \\ a_{2,3} \end{pmatrix} = a_{11}a_{21} + a_{12}a_{22} + a_{13}a_{23},$$

while for $n \times 1$ column matrices over $\mathbb{C}$ it is its complex generalization,

$$\langle a_1 | a_2 \rangle = \sum_{i=1}^n a_{1,i}^* a_{2,i}.$$

For $n \times n$ square matrices, the natural inner product uses the trace,

$$\langle a_1 | a_2 \rangle = \text{Tr}[\tilde{a}_1^\dagger \tilde{a}_2], \quad \tilde{a}^\dagger \equiv (\tilde{a}^*)^T,$$

where $\tilde{a}^\dagger$ denotes the *conjugate transpose*, or *adjoint*, of the matrix $\tilde{a}$ — we will meet this same conjugate-transpose operation again, in its full generality as the adjoint of an operator, once we discuss Hermitian operators in the next section. Finally, for square-integrable functions with $|a_1\rangle \rightarrow f(x)$ and $|a_2\rangle \rightarrow g(x)$, the inner product is

$$\langle a_1 | a_2 \rangle \equiv \int_a^b f^*(x) g(x) dx.$$

2.6 Norm of a Vector

The inner product furnishes a notion of length: the *norm* of a vector $|a\rangle$ is defined as

$$\| |a\rangle \| = \sqrt{\langle a | a \rangle},$$

and it satisfies the properties one expects of a length,

- (1) $\| |a\rangle \| \geq 0$, and $\| |a\rangle \| = 0$ iff $|a\rangle = 0$,
- (2) $\| c |a\rangle \| = |c| \| |a\rangle \|$ for $c \in \mathbb{C}$,
- (3) $\| |a_1\rangle + |a_2\rangle \| \leq \| |a_1\rangle \| + \| |a_2\rangle \|$ (the triangle inequality).

A vector is said to be *normalized* if $\| |a\rangle \| = 1$, and physical states will always be represented by normalized kets.

2.7 Orthogonality and Orthonormality

Two vectors $|a_1\rangle, |a_2\rangle$ are **orthogonal** if $\langle a_1 | a_2 \rangle = 0$, and a set of vectors $\{|a_1\rangle, |a_2\rangle, \dots, |a_n\rangle\}$ is **orthonormal** if $\langle a_i | a_j \rangle = \delta_{ij}$. Orthonormal bases are by far the most convenient to work with, because the expansion coefficients can then be read off directly by taking an inner product: if $B = \{|b_1\rangle, \dots, |b_n\rangle\}$ is an orthonormal basis, any $|a\rangle \in \alpha$ can be expanded as $|a\rangle = \sum_{i=1}^n c_i |b_i\rangle$ with $c_i \in \mathbb{C}$, and

$$\langle b_j | a \rangle = \sum_{i=1}^n c_i \langle b_j | b_i \rangle = \sum_{i=1}^n c_i \delta_{ji} = c_j,$$

giving the fundamental formula for the expansion coefficients,

$$c_i = \langle b_i | a \rangle.$$

In ordinary three-dimensional space the standard basis $|a_1\rangle = (1, 0, 0)^T, |a_2\rangle = (0, 1, 0)^T, |a_3\rangle = (0, 0, 1)^T$ is orthonormal, and for $|a\rangle = (a_x, a_y, a_z)^T$ with $a_x, a_y, a_z \in \mathbb{R}$ the norm formula reduces

to $\| |a\rangle \| = \sqrt{a_x^2 + a_y^2 + a_z^2}$; likewise the standard basis of the $n \times 1$ column matrices over $\mathbb{C}$ is orthonormal, with $\| |a\rangle \| = \sqrt{\sum_{i=1}^n |a_i|^2}$.

The square-integrable functions again require a little more care. Solutions of the time-independent Schrödinger equation for a given system can be taken as a basis for the corresponding space of wavefunctions; for a particle in a one-dimensional box of length L , for instance, the set

$$B = \left\{ |b_i\rangle \implies \sqrt{\frac{2}{L}} \sin \frac{i\pi x}{L} : i = 1, 2, \dots, \infty \right\}$$

is orthonormal and serves as a basis for wavefunctions on $[0, L]$. Here B is a *countably infinite* set, and the statement that it is complete — that every square-integrable function on $[0, L]$ really can be expanded in it, with no vectors missing — is a substantially deeper fact than in the finite-dimensional examples above: it is the statement that the particle-in-a-box eigenfunctions form a complete orthonormal set, which follows from Sturm–Liouville theory applied to the Hermitian operator $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2}$ with the box boundary conditions. We will assume the completeness of such eigenbases throughout the course; the general notion of completeness of a basis, valid equally in the finite-dimensional case, is made precise below.

2.8 Gram–Schmidt Orthogonalization

Gram–Schmidt is a prescription for constructing an orthogonal set of vectors from a set of non-orthogonal (but linearly independent) ones. Given $S = \{|a_1\rangle, |a_2\rangle, \dots, |a_n\rangle\}$ with $\langle a_i | a_j \rangle \neq 0$ in general, one constructs an orthogonal set $S' = \{|b_1\rangle, |b_2\rangle, \dots, |b_n\rangle\}$ by subtracting off, at each stage, the components already accounted for by the previous vectors:

$$\begin{aligned} |b_1\rangle &= |a_1\rangle, \\ |b_2\rangle &= |a_2\rangle - \frac{\langle b_1 | a_2 \rangle}{\langle b_1 | b_1 \rangle} |b_1\rangle, \\ |b_3\rangle &= |a_3\rangle - \frac{\langle b_2 | a_3 \rangle}{\langle b_2 | b_2 \rangle} |b_2\rangle - \frac{\langle b_1 | a_3 \rangle}{\langle b_1 | b_1 \rangle} |b_1\rangle, \\ &\vdots \\ |b_n\rangle &= |a_n\rangle - \sum_{i=1}^{n-1} \frac{\langle b_i | a_n \rangle}{\langle b_i | b_i \rangle} |b_i\rangle. \end{aligned}$$

Each $|b_k\rangle$ may then be normalized, $|\hat{b}_k\rangle = |b_k\rangle / \| |b_k\rangle \|$, to obtain an orthonormal set. As an illustration, take $|a_1\rangle = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$, $|a_2\rangle = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$, $|a_3\rangle = \begin{pmatrix} 3 \\ -7 \\ 1 \end{pmatrix}$, which are linearly independent but not orthogonal. Applying the prescription above gives

$$|b_1\rangle = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad |b_2\rangle = \begin{pmatrix} -1/2 \\ 0 \\ -1/2 \end{pmatrix}, \quad |b_3\rangle = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

(the last one up to the overall scale 3), and one may check directly that $\langle b_1 | b_2 \rangle = \langle b_1 | b_3 \rangle = \langle b_2 | b_3 \rangle = 0$, as required.

2.9 Completeness of a Basis

We have already used, without dwelling on it, the fact that a basis lets us decompose any vector into components. It is worth extracting the algebraic content of that statement, because it

will lead us directly to the notion of an operator. Let $B = \{|b_1\rangle, \dots, |b_n\rangle\}$ be orthonormal, $\langle b_i | b_j \rangle = \delta_{ij}$. For any $|a\rangle \in \alpha$,

$$|a\rangle = \sum_{i=1}^n c_i |b_i\rangle = \sum_{i=1}^n \langle b_i | a \rangle |b_i\rangle = \sum_{i=1}^n |b_i\rangle \langle b_i | a \rangle = \left[\sum_{i=1}^n |b_i\rangle \langle b_i| \right] |a\rangle.$$

The object in square brackets is something that acts on $|a\rangle$ and leaves it entirely unchanged: if $|a\rangle$ is a column matrix it acts as the identity matrix, and if $|a\rangle$ is a square-integrable function it acts as multiplication by the scalar 1. This motivates writing

$$\sum_{i=1}^n |b_i\rangle \langle b_i| = \hat{\mathbb{I}},$$

and a basis B that obeys this *completeness relation* is said to be **complete**. This is our first encounter with an object built purely out of kets and bras via the combination $|b_i\rangle \langle b_i|$ — an *operator*. The completeness relation is in fact the single most useful identity in all of quantum mechanics: it lets us insert a “resolution of the identity” $\sum_i |b_i\rangle \langle b_i|$ anywhere we like inside an expression without changing its value, a trick we will use repeatedly from here on. It is exactly this construction — linear combinations of outer products $|b_i\rangle \langle b_j|$ — that the next section develops in full generality.

3 Operators

3.1 Linear and Antilinear Operators

An operator $\hat{O}$ maps one vector to another,

$$|a_1\rangle \longrightarrow |a_2\rangle, \quad |a_2\rangle = \hat{O} |a_1\rangle, \quad |a_1\rangle, |a_2\rangle \in \alpha.$$

An operator is *linear* if it satisfies

$$\hat{O} \sum_{i=1}^n c_i |b_i\rangle = \sum_{i=1}^n c_i \hat{O} |b_i\rangle,$$

that is, if it commutes with the operation of forming a linear combination. A weaker notion, *antilinearity*, arises when the operator instead complex-conjugates the coefficients as it passes through them,

$$\hat{O} \sum_{i=1}^n c_i |b_i\rangle = \sum_{i=1}^n c_i^* \hat{O} |b_i\rangle.$$

We deal exclusively with *linear* operators in quantum mechanics; every operator appearing below, unless stated otherwise, is understood to be linear. The standard example of an antilinear operator is complex conjugation, $\hat{C}$, defined by

$$\hat{C} |a\rangle = \hat{C} \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} a_1^* \\ a_2^* \\ \vdots \\ a_n^* \end{pmatrix}.$$

Indeed, for $|a\rangle = c_1 |a_1\rangle + c_2 |a_2\rangle$,

$$\hat{C} |a\rangle = \begin{pmatrix} c_1^* a_{11}^* + c_2^* a_{21}^* \\ \vdots \\ c_1^* a_{1n}^* + c_2^* a_{2n}^* \end{pmatrix} = c_1^* \hat{C} |a_1\rangle + c_2^* \hat{C} |a_2\rangle,$$

confirming that $\hat{C}$ is antilinear rather than linear.

3.2 The Outer Product as an Operator

We have already met the outer product $|a_2\rangle\langle a_1|$ in passing; we can now recognize it as an operator in its own right, since acting on a ket with it produces another ket:

$$(|a_2\rangle\langle a_1|) |a_1\rangle = |a_2\rangle.$$

In matrix form, $|a_2\rangle\langle a_1|$ is the $n \times n$ matrix

$$\begin{pmatrix} a_{21} \\ a_{22} \\ \vdots \\ a_{2n} \end{pmatrix} \begin{pmatrix} a_{11}^* & a_{12}^* & \cdots & a_{1n}^* \end{pmatrix} = \begin{pmatrix} a_{21}a_{11}^* & a_{21}a_{12}^* & \cdots & a_{21}a_{1n}^* \\ a_{22}a_{11}^* & a_{22}a_{12}^* & \cdots & a_{22}a_{1n}^* \\ \vdots & \vdots & \ddots & \vdots \\ a_{2n}a_{11}^* & a_{2n}a_{12}^* & \cdots & a_{2n}a_{1n}^* \end{pmatrix},$$

obtained by multiplying the column vector representing $|a_2\rangle$ against the row vector representing $\langle a_1|$.

3.3 Matrix Representation of a Linear Operator

Fix an orthonormal basis $B = \{|b_1\rangle, \dots, |b_n\rangle\}$, $\langle b_i|b_j\rangle = \delta_{ij}$. Let $|a_2\rangle = \hat{O}|a_1\rangle$ with $|a_1\rangle = \sum_i c_{1,i} |b_i\rangle$, $|a_2\rangle = \sum_i c_{2,i} |b_i\rangle$, $c_{1,i}, c_{2,i} \in \mathbb{C}$. Since $\hat{O}$ is linear,

$$\sum_i c_{2,i} |b_i\rangle = \sum_i c_{1,i} \hat{O} |b_i\rangle \implies \sum_i c_{2,i} \langle b_j|b_i\rangle = \sum_i c_{1,i} \langle b_j|\hat{O}|b_i\rangle \implies c_{2,j} = \sum_i c_{1,i} \tilde{O}_{j,i},$$

where $\tilde{O}_{j,i} \equiv \langle b_j|\hat{O}|b_i\rangle \in \mathbb{C}$ is the *matrix element* of $\hat{O}$ in the basis B . Writing this out for every j ,

$$c_{2,1} = c_{1,1}\tilde{O}_{1,1} + c_{1,2}\tilde{O}_{1,2} + \cdots + c_{1,n}\tilde{O}_{1,n}, \quad \dots$$

we recognize this as ordinary matrix multiplication: every linear operator $\hat{O}$ on an n -dimensional space is represented, once a basis is fixed, by an $n \times n$ matrix $\tilde{O}$ acting on the column vectors representing kets,

$$\tilde{c}_2 = \tilde{O} \tilde{c}_1, \quad \tilde{c}_2 = \begin{pmatrix} c_{2,1} \\ \vdots \\ c_{2,n} \end{pmatrix} \text{ (represents } |a_2\rangle), \quad \tilde{c}_1 = \begin{pmatrix} c_{1,1} \\ \vdots \\ c_{1,n} \end{pmatrix} \text{ (represents } |a_1\rangle),$$

$$\tilde{O} = \begin{pmatrix} \tilde{O}_{1,1} & \tilde{O}_{1,2} & \cdots & \tilde{O}_{1,n} \\ \vdots & & \ddots & \vdots \\ \tilde{O}_{n,1} & \tilde{O}_{n,2} & \cdots & \tilde{O}_{n,n} \end{pmatrix} \quad (n \times n, \text{ representing } \hat{O}).$$

This dictionary between abstract operators and matrices, exactly parallel to the one already drawn between abstract kets and column vectors, is what lets us do all operator algebra — sums, products, inverses — as ordinary matrix algebra once a basis has been chosen.

A handful of special operators recur throughout the theory and are worth naming once and for all. The *null operator* is the one whose matrix elements all vanish, $\tilde{O}_{ij} = 0$ for every i, j ; the *identity* $\hat{\mathbb{I}}$ is the one whose matrix elements are $\tilde{O}_{ij} = \delta_{ij}$; two operators $\hat{O}_1, \hat{O}_2$ are *equal* if $\hat{O}_1|a\rangle = \hat{O}_2|a\rangle$ for every $|a\rangle \in \alpha$; they *commute* if $\hat{O}_1\hat{O}_2 - \hat{O}_2\hat{O}_1 = 0$, written $[\hat{O}_1, \hat{O}_2] = 0$, and *anticommute* if $\hat{O}_1\hat{O}_2 + \hat{O}_2\hat{O}_1 = 0$, written $\{\hat{O}_1, \hat{O}_2\} = 0$; and, if $\hat{O}_1|a_1\rangle = |a_2\rangle$ and $\hat{O}_2|a_2\rangle = |a_1\rangle$ for every such pair, then $\hat{O}_2$ is the *inverse* of $\hat{O}_1$, written $\hat{O}_2 = \hat{O}_1^{-1}$.

3.4 Matrix Element of a Product of Two Operators

With $B = \{|b_1\rangle, \dots, |b_n\rangle\}$ orthonormal and $\hat{O} = \hat{O}_1 \hat{O}_2$,

$$\begin{aligned}\tilde{O}_{ij} &= \langle b_i | \hat{O}_1 \hat{O}_2 | b_j \rangle = \langle b_i | \hat{O}_1 \left[\sum_{k=1}^n |b_k\rangle \langle b_k| \right] \hat{O}_2 | b_j \rangle \\ &= \sum_{k=1}^n \langle b_i | \hat{O}_1 | b_k \rangle \langle b_k | \hat{O}_2 | b_j \rangle = \sum_{k=1}^n (\tilde{O}_1)_{ik} (\tilde{O}_2)_{kj},\end{aligned}$$

i.e. $\tilde{O} = \tilde{O}_1 \tilde{O}_2$, ordinary matrix multiplication. The step that made this work — inserting $\sum_k |b_k\rangle \langle b_k|$ between $\hat{O}_1$ and $\hat{O}_2$ — is exactly the completeness relation of section 2.9, and the same trick of inserting a resolution of the identity wherever convenient will recur constantly from here on.

3.5 Hermitian and Unitary Operators

Given an operator $\hat{O}$, its **Hermitian conjugate**, or **adjoint**, $\hat{O}^\dagger$ is the (unique) operator defined by requiring

$$\langle a_2 | \hat{O}^\dagger | a_1 \rangle = \langle a_1 | \hat{O} | a_2 \rangle^*$$

for every pair of kets $|a_1\rangle, |a_2\rangle$ — i.e. taking the adjoint reverses the order of the bra and the ket *and* complex-conjugates the resulting number, which is exactly the matrix-level conjugate-transpose rule met in section 2.9 promoted to an operator identity. When $\hat{O}^\dagger = \hat{O}$, the operator is called **Hermitian** (or self-adjoint). In terms of matrix elements in an orthonormal basis B , the defining relation above gives directly

$$\tilde{O}_{ij}^\dagger = \langle b_i | \hat{O}^\dagger | b_j \rangle = \langle b_j | \hat{O} | b_i \rangle^* = \tilde{O}_{ji}^*,$$

so that, precisely as for the finite matrices of the previous section,

$$\tilde{O}^\dagger = (\tilde{O}^*)^T.$$

The Hermitian conjugate of a product reverses the order of the factors, just as the transpose of a product of matrices does: if $\hat{O} = \hat{O}_1 \hat{O}_2$, then

$$\hat{O}^\dagger = (\hat{O}_1 \hat{O}_2)^\dagger = \hat{O}_2^\dagger \hat{O}_1^\dagger.$$

A second class of operators defined through the adjoint, and just as important as the Hermitian ones, are the **unitary** operators, satisfying

$$\hat{O}^\dagger \hat{O} = \hat{O} \hat{O}^\dagger = \hat{\mathbb{I}}.$$

Hermitian operators will turn out to represent physical observables, and unitary operators to represent symmetries and time evolution; both notions reappear constantly for the remainder of these notes.

3.6 Basis Transformation

Let $B = \{|b_1\rangle, \dots, |b_n\rangle\}$ and $B' = \{|b'_1\rangle, \dots, |b'_n\rangle\}$ be two orthonormal bases of α ($\langle b_i | b_j \rangle = \delta_{ij} = \langle b'_i | b'_j \rangle$). The same vector $|a\rangle$ has two expansions,

$$|a\rangle = \sum_{i=1}^n c_i |b_i\rangle = \sum_{i=1}^n \langle b_i | a \rangle |b_i\rangle, \quad |a\rangle = \sum_{i=1}^n c'_i |b'_i\rangle = \sum_{i=1}^n \langle b'_i | a \rangle |b'_i\rangle,$$

and it is natural to ask how the two sets of coefficients, $\{c_i\}$ and $\{c'_i\}$, are related. Expand the new basis vectors in the old one, $|b'_i\rangle = \sum_{j=1}^n d_{ji} |b_j\rangle$ with $d_{ji} = \langle b_j | b'_i \rangle$. Then

$$|a\rangle = \sum_i c'_i |b'_i\rangle = \sum_i c'_i \left(\sum_j d_{ji} |b_j\rangle \right) = \sum_j \left(\sum_i c'_i d_{ji} \right) |b_j\rangle = \sum_j c_j |b_j\rangle,$$

so that

$$c_j = \sum_{i=1}^n c'_i d_{ji}, \quad \text{i.e.} \quad \tilde{c} = \tilde{D} \tilde{c}', \quad \tilde{c}' = \tilde{D}^{-1} \tilde{c},$$

where $\tilde{D}$, the matrix of the d_{ji} , effects the change of basis. One further shows, using $\langle b'_j | b'_i \rangle = \delta_{ji}$ and $|b'_i\rangle = \sum_j d_{ji} |b_j\rangle$, that

$$\delta_{ji} = \sum_k d_{kj}^* d_{ki} = (\tilde{D}^\dagger \tilde{D})_{ji} \implies \tilde{D}^\dagger \tilde{D} = \tilde{D} \tilde{D}^\dagger = \mathbb{I},$$

so that $\tilde{D}$ is necessarily unitary — a first hint of how naturally unitary operators arise, quite independently of the definition just given for them, whenever one passes between two orthonormal bases.

Now consider a linear operator $\hat{O}$ and ask how its matrix elements transform under this same change of basis. With $\tilde{O}'_{ij} \equiv \langle b'_i | \hat{O} | b'_j \rangle$,

$$\begin{aligned} \tilde{O}'_{ij} &= \langle b'_i | \hat{O} | b'_j \rangle = \langle b'_i | \left(\sum_{k=1}^n |b_k\rangle \langle b_k| \right) \hat{O} \left(\sum_{\ell=1}^n |b_\ell\rangle \langle b_\ell| \right) | b'_j \rangle \\ &= \sum_{k,\ell=1}^n \langle b'_i | b_k \rangle \langle b_k | \hat{O} | b_\ell \rangle \langle b_\ell | b'_j \rangle = \sum_{k,\ell=1}^n d_{ki}^* \tilde{O}_{k\ell} d_{\ell j} = (\tilde{D}^\dagger \tilde{O} \tilde{D})_{ij}. \end{aligned}$$

Hence, under a change of basis effected by the unitary matrix $\tilde{D}$,

$$\tilde{O}' = \tilde{D}^\dagger \tilde{O} \tilde{D}, \quad \hat{O}' = \hat{D}^\dagger \hat{O} \hat{D},$$

a transformation law we will see worked out concretely for the Pauli matrices in section 4.13.

3.7 Projection Operators

A *projection operator* is a linear operator that projects onto a subspace, $\hat{P} : \alpha \rightarrow \alpha' \subset \alpha$, characterized by the single algebraic condition

$$\hat{P}^2 = \hat{P}.$$

The prototype is built directly from an orthonormal basis $B = \{|b_1\rangle, \dots, |b_n\rangle\}$: let

$$\hat{P} = \sum_{i=1}^{m \leq n} |b_i\rangle \langle b_i| = \sum_{i=1}^m \hat{P}_i, \quad \hat{P}_i \equiv |b_i\rangle \langle b_i|$$

(each $\hat{P}_i$ is itself already a projection operator, onto the one-dimensional subspace spanned by $|b_i\rangle$). Then

$$\begin{aligned} \hat{P}^2 &= \left(\sum_{i=1}^m |b_i\rangle \langle b_i| \right) \left(\sum_{j=1}^m |b_j\rangle \langle b_j| \right) = \sum_{i,j=1}^m |b_i\rangle \langle b_i | b_j \rangle \langle b_j| \\ &= \sum_{i,j=1}^m |b_i\rangle \delta_{ij} \langle b_j| = \sum_{i=1}^m |b_i\rangle \langle b_i| = \hat{P}, \end{aligned}$$

confirming $\hat{P}^2 = \hat{P}$ as required. Acting on an arbitrary vector $|a\rangle = \sum_{i=1}^n c_i |b_i\rangle \in \alpha$,

$$\hat{P}|a\rangle = \left(\sum_{j=1}^m |b_j\rangle\langle b_j| \right) \sum_{i=1}^n c_i |b_i\rangle = \sum_{i=1}^n c_i \sum_{j=1}^m |b_j\rangle \delta_{ji} = \sum_{i=1}^m c_i |b_i\rangle \in \alpha' \subset \alpha,$$

so that $\hat{P}$ keeps only the components of $|a\rangle$ along the first m basis vectors and discards the rest — exactly the familiar geometric operation of projection. In ordinary three-dimensional space with the standard basis $\{\hat{x}, \hat{y}, \hat{z}\}$, for instance, taking $\hat{P} = |\hat{x}\rangle\langle\hat{x}| + |\hat{y}\rangle\langle\hat{y}|$ gives, for $|a\rangle = a_x\hat{x} + a_y\hat{y} + a_z\hat{z}$,

$$\hat{P}|a\rangle = a_x\hat{x} + a_y\hat{y} \in \alpha' = (xy\text{-plane}),$$

so that $\hat{P}$ is precisely the operation of dropping the z -component of a vector, i.e. orthogonal projection onto the xy -plane; one checks directly that $\hat{P}^2|a\rangle = \hat{P}(a_x\hat{x} + a_y\hat{y}) = a_x\hat{x} + a_y\hat{y} = \hat{P}|a\rangle$, as the general argument above guarantees.

4 Eigenvalues, Eigenvectors, and Diagonalization

4.1 The Eigenvalue Equation

The *eigenvalue equation* corresponding to an operator $\hat{O}$ is

$$\hat{O}|a\rangle = a|a\rangle,$$

where $|a\rangle$ is the **eigenvector** and a the corresponding **eigenvalue**. In a basis B , this becomes the matrix equation

$$\tilde{O}\tilde{a} = a\tilde{a} \quad (\text{for a basis } B) \implies \tilde{O}\tilde{a} = a\mathbb{I}\tilde{a} \implies (\tilde{O} - a\mathbb{I})\tilde{a} = 0,$$

which has a nontrivial solution $\tilde{a} \neq 0$ if and only if the matrix $\tilde{O} - a\mathbb{I}$ is singular, i.e. if and only if

$$\det(\tilde{O} - a\mathbb{I}) = 0 \quad (\text{the } \textit{secular} \text{ or } \textit{characteristic} \text{ equation}).$$

Expanding the determinant,

$$\det(\tilde{O} - a\mathbb{I}) = (-a)^n + \left(\sum_{i=1}^n \tilde{O}_{ii} \right) (-a)^{n-1} + \cdots + \det(\tilde{O}) a^0,$$

where $\sum_i \tilde{O}_{ii} = \text{Tr}(\tilde{O})$; this is an n th-degree polynomial in a , so the characteristic equation has exactly n roots (not necessarily distinct), counted with multiplicity, over the complex numbers.

4.2 Important Properties

Eigenvectors belonging to distinct eigenvalues are automatically linearly independent. To see this, let $\tilde{O}\tilde{a}_1 = a_1\tilde{a}_1$, $\tilde{O}\tilde{a}_2 = a_2\tilde{a}_2$, with $a_1 \neq a_2$, and suppose, for contradiction, that $\tilde{a}_1, \tilde{a}_2$ are linearly *dependent*, i.e.

$$c_1\tilde{a}_1 + c_2\tilde{a}_2 = 0, \quad c_1, c_2 \text{ not both } 0. \quad (*)$$

Applying $\tilde{O}$ to $(*)$ gives $c_1a_1\tilde{a}_1 + c_2a_2\tilde{a}_2 = 0$, while multiplying $(*)$ itself by the scalar a_2 gives $c_1a_2\tilde{a}_1 + c_2a_2\tilde{a}_2 = 0$; subtracting the second from the first eliminates the c_2 term entirely,

$$c_1(a_1 - a_2)\tilde{a}_1 = 0.$$

Since $a_1 \neq a_2$, this forces $c_1 = 0$ (note that no division by a_1 or a_2 was needed, so the argument holds even if one of the eigenvalues is zero); substituting back into $(*)$ then gives $c_2\tilde{a}_2 = 0$, and since $\tilde{a}_2 \neq 0$ this forces $c_2 = 0$ as well, contradicting the assumption that c_1, c_2 were not both zero. So $\tilde{a}_1, \tilde{a}_2$ must in fact be linearly independent. More generally, if $\tilde{O}$ has r distinct eigenvalues, it has at least r linearly independent eigenvectors. The converse, however, is false: linearly independent eigenvectors can perfectly well share the same eigenvalue. For example, $\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ has $a_1 = a_2 = 1$, yet $\tilde{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\tilde{a}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are linearly independent.

4.3 Diagonalization

Two matrices $\tilde{O}_1, \tilde{O}_2$ are called *similar* if $\tilde{O}_1 = \tilde{S}^{-1} \tilde{O}_2 \tilde{S}$ for some invertible $\tilde{S}$, and $\tilde{O}$ is *diagonalizable* if it is similar to a diagonal matrix, i.e.

$$\tilde{O} = \tilde{S} \tilde{D} \tilde{S}^{-1}, \quad \tilde{D} = \text{diag}(d_1, \dots, d_n),$$

where the d_i are the eigenvalues of $\tilde{O}$. To see when this is possible, suppose $\tilde{O}$ has n linearly independent eigenvectors $\tilde{a}_i$ ($\tilde{O}\tilde{a}_i = a_i\tilde{a}_i$, $i = 1, \dots, n$), and construct $\tilde{S} = (\tilde{a}_1 \ \tilde{a}_2 \ \dots \ \tilde{a}_n)$, the matrix whose columns are these eigenvectors. Then

$$\begin{aligned} \tilde{O} \tilde{S} &= (a_1 \tilde{a}_1 \ a_2 \tilde{a}_2 \ \dots \ a_n \tilde{a}_n) = (\tilde{a}_1 \ \dots \ \tilde{a}_n) \begin{pmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{pmatrix} = \tilde{S} \tilde{D} \\ \implies \tilde{D} &= \tilde{S}^{-1} \tilde{O} \tilde{S} \iff \tilde{O} = \tilde{S} \tilde{D} \tilde{S}^{-1}, \end{aligned}$$

so that $\tilde{O}$ is diagonalizable precisely when it possesses a full set of n linearly independent eigenvectors. (That these n linearly independent eigenvectors can always be *chosen orthonormal*, so that $\tilde{S}$ is in fact unitary, will be proved below for the important special case of a Hermitian operator.) A closely related fact, which we will put to extensive use once we discuss compatible observables, is that if $[\tilde{O}_1, \tilde{O}_2] = 0$, then $\tilde{O}_1, \tilde{O}_2$ are *simultaneously* diagonalizable: $\tilde{O}_1 = \tilde{S}^{-1} \tilde{D}_1 \tilde{S}$ and $\tilde{O}_2 = \tilde{S}^{-1} \tilde{D}_2 \tilde{S}$ for one and the same $\tilde{S}$. We return to this result, and to its converse, in full generality in section 6.4.

4.4 Normal Matrices

A matrix is called *normal* if it commutes with its own adjoint, $\tilde{O}^\dagger \tilde{O} = \tilde{O} \tilde{O}^\dagger$. Unitary matrices are automatically normal, since $\tilde{O}^\dagger \tilde{O} = \tilde{O} \tilde{O}^\dagger = \mathbb{I}$ for them, and so are Hermitian matrices, since $\tilde{O}^\dagger \tilde{O} = \tilde{O}^2 = \tilde{O} \tilde{O}^\dagger$ for them; normality is thus the common feature that makes both classes diagonalizable by a unitary transformation, as we will see explicitly in section 6.5.

4.5 Eigenvalues of Unitary Matrices Have Unit Modulus

Let $\tilde{O}^\dagger \tilde{O} = \tilde{O} \tilde{O}^\dagger = \mathbb{I}$ and $\tilde{O} \tilde{a} = a \tilde{a}$ with $\tilde{a} \neq 0$. Unitary operators preserve the norm of every vector (that is the content of $\tilde{O}^\dagger \tilde{O} = \mathbb{I}$: $\|\tilde{O} \tilde{a}\|^2 = \tilde{a}^\dagger \tilde{O}^\dagger \tilde{O} \tilde{a} = \tilde{a}^\dagger \tilde{a} = \|\tilde{a}\|^2$), so in particular

$$\|\tilde{a}\|^2 = \|\tilde{O} \tilde{a}\|^2 = \|a \tilde{a}\|^2 = a^* a \|\tilde{a}\|^2 = |a|^2 \|\tilde{a}\|^2,$$

and since $\|\tilde{a}\|^2 \neq 0$, this forces $|a|^2 = 1$. So every eigenvalue of a unitary matrix lies on the unit circle in the complex plane, $a = e^{i\theta}$ for some real θ — consistent with unitary operators representing symmetries that preserve the length of every vector.

4.6 Eigenvalue Problem of a Hermitian Operator

Let $\hat{O} = \hat{O}^\dagger$ with eigenvectors $|a_i\rangle, |a_j\rangle$,

$$\hat{O} |a_i\rangle = a_i |a_i\rangle, \quad \hat{O} |a_j\rangle = a_j |a_j\rangle.$$

Then

$$\begin{aligned} \langle a_j | \hat{O} | a_i \rangle &= a_i \langle a_j | a_i \rangle \\ \langle a_i | \hat{O} | a_j \rangle &= a_j \langle a_i | a_j \rangle \\ \implies \langle a_i | \hat{O} | a_j \rangle^* &= a_j^* \langle a_j | a_i \rangle \\ \implies \langle \hat{O} a_j | a_i \rangle &= a_j^* \langle a_j | a_i \rangle \\ \implies \langle a_j | \hat{O} | a_i \rangle &= a_j^* \langle a_j | a_i \rangle \end{aligned} \tag{1}$$

(using Hermiticity, $\hat{O}^\dagger = \hat{O}$, in the last line). Comparing (1) and (2),

$$a_i \langle a_j | a_i \rangle = a_j^* \langle a_j | a_i \rangle, \quad \text{for all } i, j.$$

Two immediate and far-reaching consequences follow. Taking $i = j$ gives $a_i = a_i^*$, so **the eigenvalues of a Hermitian operator are always real** — the algebraic fact that makes Hermitian operators the right objects to represent physical observables, whose measured values must of course be real numbers. Taking $i \neq j$ and using this reality, $(a_i - a_j) \langle a_j | a_i \rangle = 0$; hence if $a_i \neq a_j$ then $\langle a_j | a_i \rangle = 0$, so that **eigenvectors belonging to different eigenvalues of a Hermitian operator are automatically orthogonal**. Only when $\langle a_j | a_i \rangle \neq 0$ can we have $a_i = a_j$: this is the case of **degenerate eigenvalues**, for which the corresponding eigenvectors need not *a priori* be orthogonal, though of course they can always be chosen so. Putting these two facts together: a complete set of eigenvectors of a Hermitian operator can always be arranged into an orthonormal basis for the whole space — eigenvectors of distinct eigenvalues are already orthogonal, and within any degenerate eigenspace one simply starts from an arbitrary linearly independent set of eigenvectors spanning it and applies Gram–Schmidt.

Two examples illustrate the two cases. First, take $\tilde{O} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, which has three distinct eigenvalues $a_1 = -1$, $a_2 = 0$, $a_3 = 1$, with normalized eigenvectors

$$\tilde{a}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \tilde{a}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \tilde{a}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix},$$

already mutually orthogonal, exactly as guaranteed. Second, take $\tilde{O} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$, which has $a_1 = -1$, $a_2 = 1$, $a_3 = 1$, so that a_2 and a_3 are degenerate, with

$$\tilde{a}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \quad \tilde{a}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix},$$

and, by inspection, $\tilde{a}_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, which happens already to be orthogonal to $\tilde{a}_1, \tilde{a}_2$, so no Gram–Schmidt is even needed here. Had it not come out orthogonal automatically, the systematic procedure would have been to pick any trial vector independent of the two known eigenvectors, e.g. $\tilde{b}_3 = \alpha(1, 1, 1)^T$, and apply Gram–Schmidt to it against $\tilde{a}_1, \tilde{a}_2$ to obtain an eigenvector orthogonal to both — which, in this case, reproduces $\tilde{a}_3 = (1, 0, 0)^T$.

4.7 Unitary Transformations Preserve Eigenvalues

Let $\tilde{O}\tilde{a} = a\tilde{a}$ and let $\tilde{S}$ be unitary ($\tilde{S}^{-1}\tilde{S} = \mathbb{I} = \tilde{S}\tilde{S}^{-1}$). Then

$$\begin{aligned} \tilde{O}\mathbb{I}\tilde{a} = a\tilde{a} &\implies \tilde{O}\tilde{S}^{-1}\tilde{S}\tilde{a} = a\tilde{a} \implies (\tilde{S}\tilde{O}\tilde{S}^{-1})(\tilde{S}\tilde{a}) = a(\tilde{S}\tilde{a}) \\ &\implies (\tilde{S}\tilde{O}\tilde{S}^{-1})\tilde{a}' = a\tilde{a}', \quad \tilde{a}' \equiv \tilde{S}\tilde{a}, \end{aligned}$$

so a unitary similarity transformation of a matrix leaves its eigenvalues completely unchanged, merely transforming the eigenvector $\tilde{a} \mapsto \tilde{S}\tilde{a}$ — as it must, since a change of orthonormal basis (section 3.5) cannot alter the intrinsic, basis-independent content of the eigenvalue problem.

4.8 Determining the Diagonalizing Matrix $\tilde{S}$

Let $\tilde{S}$ diagonalize a Hermitian $\tilde{O}$ ($\tilde{O}^\dagger = \tilde{O}$), in the same convention as above, $\tilde{S}^{-1}\tilde{S} = \mathbb{I} = \tilde{S}\tilde{S}^{-1}$, $\tilde{S}^{-1} = \tilde{S}^\dagger$:

$$\tilde{S}^{-1}\tilde{O}\tilde{S} = \tilde{D} = \text{diag}(d_1, \dots, d_n), \quad \text{equivalently} \quad \tilde{O}\tilde{S} = \tilde{S}\tilde{D}.$$

Writing $\tilde{S} = (\tilde{a}_1 \tilde{a}_2 \cdots \tilde{a}_n)$ in terms of its columns, the i th column of $\tilde{O}\tilde{S} = \tilde{S}\tilde{D}$ reads $\tilde{O}\tilde{a}_i = d_i\tilde{a}_i$: each column $\tilde{a}_i$ of $\tilde{S}$ is precisely the eigenvector of $\tilde{O}$ with eigenvalue d_i , exactly as found in the previous section. In particular $\tilde{S}$ is not unique: it is fixed only up to a reshuffling of the eigenvectors (corresponding to a reordering of the d_i along the diagonal of $\tilde{D}$) and up to rescaling each eigenvector by a phase.

4.9 Spectral Decomposition of a Hermitian Operator

Written in its own eigenbasis $\{|a_i\rangle\}$, a Hermitian operator takes the strikingly simple form

$$\hat{O} = \sum_{i=1}^n a_i |a_i\rangle\langle a_i|,$$

called its *spectral decomposition*: $\hat{O}$ is nothing but a weighted sum of the projection operators onto its own eigenspaces, weighted by the corresponding eigenvalues. This representation is the natural culmination of everything said so far about Hermitian operators, and it is the form in which such operators will most often be used from now on.

4.10 Functions of an Operator

We restrict to functions expressible as a power series, $f(x) = \sum_{i=0}^{\infty} a_i x^i$, $x \in \mathbb{C}$, and *define* the same function of a Hermitian operator $\hat{O}$ by

$$f(\hat{O}) = \sum_{i=0}^{\infty} a_i \hat{O}^i,$$

provided the sum converges to a definite limit (e.g. $e^{\hat{O}} = \sum_{m=0}^{\infty} \hat{O}^m/m!$). Going to the eigenbasis $\{|a_i\rangle\}$ of $\hat{O}$ (where $\hat{O} = \text{diag}(a_1, \dots, a_n)$, so $\hat{O}^m = \text{diag}(a_1^m, \dots, a_n^m)$),

$$e^{\hat{O}} = \text{diag}\left(\sum_{m=0}^{\infty} \frac{a_1^m}{m!}, \dots, \sum_{m=0}^{\infty} \frac{a_n^m}{m!}\right) = \sum_{i=1}^n e^{a_i} |a_i\rangle\langle a_i|,$$

and the same reasoning applies to any power series, giving in general

$$f(\hat{O}) = \sum_{i=1}^n f(a_i) |a_i\rangle\langle a_i|, \quad \hat{O}|a_i\rangle = a_i|a_i\rangle, \quad i = 1, \dots, n, \quad \hat{O} \text{ Hermitian.}$$

In words: to apply f to an operator, diagonalize the operator, apply f to each eigenvalue, and reassemble the spectral decomposition. This is the rigorous meaning behind expressions such as $e^{i\hat{H}t/\hbar}$ that appear throughout quantum dynamics.

4.11 Derivative of an Operator

For an operator $\hat{O}(\lambda)$ depending on a parameter λ , the derivative is defined exactly as for an ordinary function,

$$\frac{d\hat{O}(\lambda)}{d\lambda} = \lim_{\Delta\lambda \rightarrow 0} \frac{\hat{O}(\lambda + \Delta\lambda) - \hat{O}(\lambda)}{\Delta\lambda}.$$

If $\hat{A}(\lambda) \equiv d\hat{O}(\lambda)/d\lambda$ and $\hat{O}(\lambda)$ has matrix representation $\tilde{O}(\lambda)$ in a fixed (λ -independent) basis, then $[\tilde{A}(\lambda)]_{ij} = d[\tilde{O}(\lambda)]_{ij}/d\lambda$ in the same basis: differentiating an operator is the same as differentiating each of its matrix elements. As an illustration, take $\hat{O}(\lambda) = e^{\lambda\hat{A}}$ with $\hat{A}$ Hermitian and independent of λ ; then

$$\frac{d\hat{O}(\lambda)}{d\lambda} = \hat{A}e^{\lambda\hat{A}} = e^{\lambda\hat{A}}\hat{A} = \hat{O}(\lambda)\hat{A},$$

since $\hat{A}$ commutes with any power of itself, so that in this particular (but very common) case the operator and its derivative simply commute with each other — a fact used repeatedly once we come to solve the Schrödinger equation.

4.12 Example: The Rotation Operator

Consider ordinary 3D space with basis $|b_1\rangle = \hat{x}, |b_2\rangle = \hat{y}, |b_3\rangle = \hat{z}$. A rotation by $\pi/2$ about $\hat{z}$ (counter-clockwise) sends $\hat{z}' = \hat{z}, \hat{x}' = \hat{y}, \hat{y}' = -\hat{x}$, i.e. acting on column vectors, $\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \tilde{R} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ with $\tilde{R}_{ij} = \langle b_i | \hat{R} | b_j \rangle$. Explicitly $\tilde{R}_{11} = \langle b_1 | b_2 \rangle = 0$, $\tilde{R}_{12} = -\langle b_1 | b_1 \rangle = -1$, $\tilde{R}_{13} = \langle b_1 | b_3 \rangle = 0$, and so on, giving

$$\tilde{R} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

This is the special case $\theta = \pi/2$ of the general rotation-about- $\hat{z}$ matrix

$$\tilde{R}_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which is unitary (in fact real orthogonal, $\tilde{R}_z^T \tilde{R}_z = \mathbb{I}$) for every θ , exactly as it must be, since rotations preserve lengths and angles. Writing $\tilde{R}_z(\theta) = e^{-i\theta\hat{J}_z}$ for a suitable generator $\hat{J}_z$ is precisely the “function of an operator” and “derivative of an operator” machinery of the preceding two sections, applied to the generator of rotations. Differentiating at $\theta = 0$ gives $\hat{J}_z = i \left. \frac{d\tilde{R}_z}{d\theta} \right|_0 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, which is Hermitian (as it must be, for $e^{-i\theta\hat{J}_z}$ to be unitary for every real θ by the functions-of-an-operator machinery above) — the same construction that reappears, with $\hat{J}_z \rightarrow \hat{S}_z$, when we come to build the spin operators later in the course.

4.13 Example: The Pauli Matrices

$$\begin{aligned} \tilde{\sigma}_1 = \tilde{\sigma}_x &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, & \tilde{\sigma}_2 = \tilde{\sigma}_y &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \\ \tilde{\sigma}_3 = \tilde{\sigma}_z &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, & \tilde{\sigma}_0 = \mathbb{I} &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Each $\tilde{\sigma}_i$ ($i = 1, 2, 3$) is Hermitian, unitary, and satisfies $\tilde{\sigma}_i^2 = \mathbb{I}$. Their eigenvalues are $\lambda_1 = +1, \lambda_2 = -1$ for every i , with eigenvectors $\hat{\sigma}_i |1\rangle = \lambda_1 |1\rangle, \hat{\sigma}_i |2\rangle = \lambda_2 |2\rangle$:

$i = 1$	$i = 2$	$i = 3$
$ 1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$ 1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$	$ 1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
$ 2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$	$ 2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$	$ 2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Pauli matrices are usually written in the eigenbasis of $\hat{\sigma}_3$; in its *own* eigenbasis, of course, every Pauli matrix takes the same diagonal form $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, in keeping with the spectral decomposition of the previous section. The (unitary, Hermitian) matrix transforming the basis from the $\hat{\sigma}_1$ -eigenbasis to the $\hat{\sigma}_3$ -eigenbasis, and vice versa, is the **Hadamard matrix**

$$\tilde{S} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{\tilde{\sigma}_1 + \tilde{\sigma}_3}{\sqrt{2}},$$

an explicit instance of the basis-transformation law $\tilde{O}' = \tilde{D}^\dagger \tilde{O} \tilde{D}$ of section 3.5. Finally, the most general Hermitian operator on the 2-dimensional space spanned by $\{|1\rangle, |2\rangle\}$ can always be written as

$$\hat{O} = \frac{1}{2} \sum_{i=0}^3 c_i \hat{\sigma}_i, \quad c_i \in \mathbb{R},$$

a formula worth dwelling on: since $\{\tilde{\sigma}_0, \tilde{\sigma}_1, \tilde{\sigma}_2, \tilde{\sigma}_3\}$ are four linearly independent 2×2 Hermitian matrices, they form a basis for *all* 2×2 Hermitian matrices (a four-real-dimensional space), which is exactly why every such operator can be written in this form with real coefficients c_i . This is the algebraic fact underlying the Bloch-sphere picture of a two-level (spin- $\frac{1}{2}$) system that will be used heavily once we discuss spin.

5 Hilbert Space

We are now in a position to name, precisely, the structure that has been under construction since section 2.9. A *Hilbert space* is a linear vector space on which an inner product is defined, i.e. an **inner-product space**, with one further technical requirement: *completeness*, meaning that every Cauchy sequence of vectors in the space converges to a vector that is also in the space. (This is a different, though related, use of the word “complete” from the completeness of a *basis* discussed in section 2.9 — context always makes clear which is meant.) For finite-dimensional spaces such as $\mathbb{C}^n$, completeness in this sense is automatic and imposes no extra condition at all; for infinite-dimensional spaces, such as the space of square-integrable functions $L^2[a, b]$ that wavefunctions live in, it is a genuine extra condition, and it is precisely why square-integrable functions — rather than, say, all differentiable functions, or all continuous functions — form the right arena for quantum mechanics. The dimension of a Hilbert space can be finite, as for a spin degree of freedom, or (countably or uncountably) infinite, as for a particle moving in continuous space; both cases will be needed before these notes are through.

With this, we have assembled exactly the mathematical structure anticipated at the outset: physical states of a quantum system are rays in a complex Hilbert space $\mathcal{H}$; physical observables (such as μ_z in the Stern–Gerlach experiment) are represented by Hermitian operators on $\mathcal{H}$; the possible outcomes of a measurement of an observable are the eigenvalues of the corresponding operator, guaranteed real by the Hermitian eigenvalue problem of section 3.5; and the sequential-measurement puzzle raised at the very beginning of these notes is resolved once one accepts that measuring one observable ($\hat{\sigma}_x$, say) generically changes the state’s expansion in the eigenbasis of another, non-commuting observable ($\hat{\sigma}_z$) — exactly analogous to the polarizer example, where inserting a filter X' between X and Y changes what Y subsequently sees. Everything that follows

in the rest of these notes — continuous bases, composite systems, the uncertainty principle, and compatible observables — is built entirely within this same Hilbert-space framework, extending the finite-dimensional machinery developed so far to the settings quantum mechanics actually requires.

6 Further Topics

The machinery built up over the last three sections — linear vector spaces, operators, eigenvalue problems, and Hilbert space — already suffices to state the postulates of quantum mechanics for a system with a finite number of degrees of freedom. Four extensions are needed before the formalism is ready for genuine physical use: bases labeled by a continuous rather than discrete index (needed as soon as position or momentum is treated quantum mechanically), the tensor product construction that describes systems built out of several parts, the uncertainty relation that quantifies the price paid when two observables fail to commute, and the notion of a complete set of commuting observables that tells us exactly how many labels are needed to specify a quantum state uniquely. Along the way it is also worth completing the discussion of diagonalization from section 4.13 by identifying the most general class of operators that admit an orthonormal eigenbasis. We take up each of these in turn, using throughout exactly the same notation and conventions established above.

6.1 Continuous Bases

Every basis considered so far has been discrete: a finite (or, for the particle in a box, countably infinite) list of basis kets $|b_i\rangle$ labeled by an integer i . Position in space, however, takes a continuum of values, and it is natural to ask whether an operator such as $\hat{x}$, with $\hat{x}|x\rangle = x|x\rangle$ for every real number x , can be fit into the same bra-ket language. It can, provided two of our discrete-basis formulas are suitably reinterpreted.

The orthonormality relation $\langle b_i|b_j\rangle = \delta_{ij}$ must become a statement about a continuous label. The right replacement for the Kronecker delta is the *Dirac delta function* $\delta(x - x')$, defined by the property

$$\int_{-\infty}^{\infty} dx' f(x') \delta(x - x') = f(x)$$

for any sufficiently well-behaved function f , and we *postulate* the orthonormality of the position eigenkets in the form

$$\langle x|x'\rangle = \delta(x - x').$$

Unlike a discrete basis ket, $|x\rangle$ is not normalizable ($\langle x|x\rangle = \delta(0)$ is formally infinite) and so does not represent a physically realizable state by itself; it is, nonetheless, an extremely useful idealization, exactly as a discrete eigenbasis is useful even though individual basis vectors need not correspond to anything measured directly.

The completeness relation $\hat{\mathbb{I}} = \sum_i |b_i\rangle\langle b_i|$ of section 2.9 becomes, with the sum over the continuous label x replaced by an integral,

$$\hat{\mathbb{I}} = \int_{-\infty}^{\infty} dx |x\rangle\langle x|.$$

That this is the correct completeness relation is checked exactly as before: acting with it on $|x'\rangle$ reproduces $|x'\rangle$ itself,

$$\left(\int dx |x\rangle\langle x| \right) |x'\rangle = \int dx |x\rangle \langle x|x'\rangle = \int dx |x\rangle \delta(x - x') = |x'\rangle,$$

using the defining property of the delta function in the last step. Given this completeness relation, an arbitrary state $|\psi\rangle$ expands as

$$|\psi\rangle = \hat{\mathbb{I}}|\psi\rangle = \int dx |x\rangle \langle x|\psi\rangle = \int dx \psi(x) |x\rangle, \quad \psi(x) \equiv \langle x|\psi\rangle,$$

which is nothing but the familiar Schrödinger *wavefunction* $\psi(x)$, now revealed to be exactly the continuous-basis analogue of the discrete expansion coefficients $c_i = \langle b_i|a\rangle$ used throughout section 2.9: $\psi(x)$ is simply the component of $|\psi\rangle$ along the continuous basis ket $|x\rangle$. The inner product of two states follows the same pattern as section 2.9's discrete formula, with the sum over i replaced by an integral over x ,

$$\langle\phi|\psi\rangle = \langle\phi|\left(\int dx |x\rangle\langle x|\right)|\psi\rangle = \int dx \langle\phi|x\rangle \langle x|\psi\rangle = \int dx \phi^*(x) \psi(x),$$

and in particular the normalization condition $\langle\psi|\psi\rangle = 1$ becomes the familiar statement that $|\psi(x)|^2$ integrates to one, i.e. is a probability density.

Momentum admits exactly the same treatment: momentum eigenkets $|p\rangle$ satisfy $\hat{p}|p\rangle = p|p\rangle$, with orthonormality $\langle p|p'\rangle = \delta(p - p')$ and completeness $\hat{\mathbb{I}} = \int dp |p\rangle\langle p|$, and an arbitrary state expands equally well in the momentum basis, $|\psi\rangle = \int dp \tilde{\psi}(p) |p\rangle$, with $\tilde{\psi}(p) \equiv \langle p|\psi\rangle$ the momentum-space wavefunction. Since $|\psi\rangle$ is one and the same abstract vector regardless of which basis is used to expand it, $\psi(x)$ and $\tilde{\psi}(p)$ must be related, and the relation is fixed entirely by the overlap $\langle x|p\rangle$ between the two continuous bases, which one shows (from the fact that $\hat{p}$ generates translations of $\hat{x}$) takes the form

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}.$$

As a check, this overlap correctly reproduces the position-space completeness relation via the momentum basis: inserting $\hat{\mathbb{I}} = \int dp |p\rangle\langle p|$ between $\langle x|$ and $|x'\rangle$,

$$\langle x|x'\rangle = \int dp \langle x|p\rangle \langle p|x'\rangle = \frac{1}{2\pi\hbar} \int dp e^{ip(x-x')/\hbar} = \delta(x - x'),$$

using the Fourier representation of the Dirac delta function in the last step, exactly as required by the postulated orthonormality of $|x\rangle$. Inserting the same completeness relation into $\tilde{\psi}(p) = \langle p|\psi\rangle$ gives the familiar Fourier-transform relation between the position- and momentum-space wavefunctions,

$$\tilde{\psi}(p) = \langle p|\psi\rangle = \int dx \langle p|x\rangle \langle x|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int dx e^{-ipx/\hbar} \psi(x),$$

so that $\psi(x)$ and $\tilde{\psi}(p)$ are simply the same state $|\psi\rangle$ written out in two different (continuous) bases, in precise analogy with the discrete change-of-basis formula of section 3.5.

6.2 Composite Systems and Tensor Product Spaces

Everything so far has described a single system. Quantum mechanics must also describe *composite* systems — two particles, or a single particle's position together with its spin — and the mathematical structure that combines two Hilbert spaces into one is the *tensor product*.

Let $\mathcal{H}_1$ be an n_1 -dimensional Hilbert space with orthonormal basis $\{|u_i\rangle\}_{i=1}^{n_1}$, and $\mathcal{H}_2$ an n_2 -dimensional Hilbert space with orthonormal basis $\{|v_j\rangle\}_{j=1}^{n_2}$. Their tensor product $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ is the $(n_1 n_2)$ -dimensional Hilbert space spanned by the $n_1 n_2$ product basis kets

$$|u_i\rangle \otimes |v_j\rangle \equiv |u_i\rangle |v_j\rangle \equiv |u_i, v_j\rangle,$$

one for every pair (i, j) . The tensor product of a general vector $|a\rangle = \sum_i c_i |u_i\rangle \in \mathcal{H}_1$ with a general vector $|b\rangle = \sum_j d_j |v_j\rangle \in \mathcal{H}_2$ is defined to be *bilinear*,

$$|a\rangle \otimes |b\rangle \equiv \left(\sum_i c_i |u_i\rangle \right) \otimes \left(\sum_j d_j |v_j\rangle \right) = \sum_{i,j} c_i d_j |u_i, v_j\rangle,$$

so that $\otimes$ distributes over addition in each slot separately, $(\alpha |a_1\rangle + \beta |a_2\rangle) \otimes |b\rangle = \alpha(|a_1\rangle \otimes |b\rangle) + \beta(|a_2\rangle \otimes |b\rangle)$, and similarly in the second slot. The inner product on $\mathcal{H}$ is likewise built factor by factor,

$$(\langle a_1| \otimes \langle b_1|)(|a_2\rangle \otimes |b_2\rangle) = \langle a_1|a_2\rangle \langle b_1|b_2\rangle,$$

which in particular makes the product basis $\{|u_i, v_j\rangle\}$ orthonormal in $\mathcal{H}$ whenever $\{|u_i\rangle\}$ and $\{|v_j\rangle\}$ are orthonormal in $\mathcal{H}_1, \mathcal{H}_2$. Operators acting on one factor alone extend to $\mathcal{H}$ in the obvious way, $\hat{A} \otimes \hat{B}$ being defined on product vectors by

$$(\hat{A} \otimes \hat{B})(|a\rangle \otimes |b\rangle) = (\hat{A}|a\rangle) \otimes (\hat{B}|b\rangle),$$

and extended to all of $\mathcal{H}$ by linearity; an operator such as $\hat{A}_1$ acting on system 1 alone, leaving system 2 untouched, is simply $\hat{A}_1 \otimes \hat{\mathbb{I}}_2$.

A general vector of $\mathcal{H}_1 \otimes \mathcal{H}_2$ is not, however, required to be a single product $|a\rangle \otimes |b\rangle$: the most general element of $\mathcal{H}$ is a linear combination

$$|\Psi\rangle = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} c_{ij} |u_i, v_j\rangle,$$

for an arbitrary array of coefficients c_{ij} , and $|\Psi\rangle$ is a simple product state, $|\Psi\rangle = |a\rangle \otimes |b\rangle$, if and only if the matrix (c_{ij}) factorizes as $c_{ij} = \alpha_i \beta_j$ (equivalently, has rank one). A composite-system state that cannot be written as a product of a state of system 1 with a state of system 2 is called *entangled*, and the possibility of entanglement — with no analogue anywhere in classical mechanics — is exactly what the tensor product, rather than some simpler combination of $\mathcal{H}_1$ and $\mathcal{H}_2$, makes room for.

The point is best seen in the smallest nontrivial example: two spin- $\frac{1}{2}$ particles, each with its own two-dimensional Hilbert space $\mathcal{H}_i = \mathbb{C}^2$ spanned by the $\hat{\sigma}_z$ -eigenstates $|\uparrow\rangle, |\downarrow\rangle$ of section 4.13. The composite space $\mathcal{H}_1 \otimes \mathcal{H}_2 \cong \mathbb{C}^4$ has product basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$, using the shorthand $|s_1 s_2\rangle \equiv |s_1\rangle \otimes |s_2\rangle$. Consider the particular state

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle),$$

known as the *singlet state*. Suppose, for contradiction, that $|\Psi^-\rangle$ were a product state, $|\Psi^-\rangle = (\alpha |\uparrow\rangle + \beta |\downarrow\rangle) \otimes (\gamma |\uparrow\rangle + \delta |\downarrow\rangle)$ for some $\alpha, \beta, \gamma, \delta \in \mathbb{C}$. Expanding the right-hand side,

$$(\alpha |\uparrow\rangle + \beta |\downarrow\rangle) \otimes (\gamma |\uparrow\rangle + \delta |\downarrow\rangle) = \alpha\gamma |\uparrow\uparrow\rangle + \alpha\delta |\uparrow\downarrow\rangle + \beta\gamma |\downarrow\uparrow\rangle + \beta\delta |\downarrow\downarrow\rangle,$$

and comparing coefficients of the four (orthonormal, hence linearly independent) product basis kets against $|\Psi^-\rangle$ forces

$$\alpha\gamma = 0, \quad \alpha\delta = \frac{1}{\sqrt{2}}, \quad \beta\gamma = -\frac{1}{\sqrt{2}}, \quad \beta\delta = 0.$$

From $\alpha\gamma = 0$, either $\alpha = 0$ or $\gamma = 0$. If $\alpha = 0$, then $\alpha\delta = 0$, contradicting $\alpha\delta = 1/\sqrt{2}$. If instead $\gamma = 0$, then $\beta\gamma = 0$, contradicting $\beta\gamma = -1/\sqrt{2}$. Either way we reach a contradiction, so no such $\alpha, \beta, \gamma, \delta$ exist: $|\Psi^-\rangle$ cannot be written as a product of a state of particle 1 with a state of particle 2. The singlet is entangled — an explicit, completely algebraic instance of a state with no classical counterpart, made possible entirely by the tensor-product structure of composite quantum systems.

6.3 The Generalized Uncertainty Principle

We return now to a tool used implicitly throughout the last two sections — the inner product's basic inequality — and put it to work extracting a genuinely new physical statement out of the algebra of Hermitian operators: the uncertainty principle.

For any two vectors $|a\rangle, |b\rangle$ in an inner-product space, the *Cauchy–Schwarz inequality* states that

$$|\langle a|b\rangle|^2 \leq \langle a|a\rangle \langle b|b\rangle.$$

The proof is a direct application of the Gram–Schmidt idea of section 2.9: decompose $|b\rangle$ into a piece along $|a\rangle$ and a piece orthogonal to it,

$$|b\rangle = \frac{\langle a|b\rangle}{\langle a|a\rangle} |a\rangle + |c\rangle, \quad |c\rangle \equiv |b\rangle - \frac{\langle a|b\rangle}{\langle a|a\rangle} |a\rangle,$$

so that $\langle a|c\rangle = \langle a|b\rangle - \frac{\langle a|b\rangle}{\langle a|a\rangle} \langle a|a\rangle = 0$ by construction. Then, using the norm's positivity, $\langle c|c\rangle \geq 0$, together with $\langle a|c\rangle = 0 = \langle c|a\rangle$,

$$0 \leq \langle c|c\rangle = \langle b|b\rangle - \frac{\langle b|a\rangle}{\langle a|a\rangle} \langle a|b\rangle - \frac{\langle a|b\rangle}{\langle a|a\rangle} \langle b|a\rangle + \frac{|\langle a|b\rangle|^2}{\langle a|a\rangle^2} \langle a|a\rangle = \langle b|b\rangle - \frac{|\langle a|b\rangle|^2}{\langle a|a\rangle},$$

which rearranges immediately to $|\langle a|b\rangle|^2 \leq \langle a|a\rangle \langle b|b\rangle$, as claimed, with equality if and only if $|c\rangle = 0$, i.e. if and only if $|a\rangle$ and $|b\rangle$ are proportional.

Now let $\hat{A}, \hat{B}$ be two Hermitian operators, and let $|\psi\rangle$ be any normalized state, $\langle\psi|\psi\rangle = 1$. Define the *deviation operators*

$$\Delta\hat{A} \equiv \hat{A} - \langle\hat{A}\rangle \hat{\mathbb{I}}, \quad \Delta\hat{B} \equiv \hat{B} - \langle\hat{B}\rangle \hat{\mathbb{I}}, \quad \langle\hat{A}\rangle \equiv \langle\psi|\hat{A}|\psi\rangle,$$

which are themselves Hermitian, being a Hermitian operator minus a real multiple of the identity. The variance of $\hat{A}$ in the state $|\psi\rangle$ is $(\Delta A)^2 \equiv \langle(\Delta\hat{A})^2\rangle = \langle\psi|(\Delta\hat{A})^2|\psi\rangle$, and similarly for $\hat{B}$. Applying the Cauchy–Schwarz inequality to the two vectors $|a\rangle \equiv \Delta\hat{A}|\psi\rangle$ and $|b\rangle \equiv \Delta\hat{B}|\psi\rangle$, whose norms squared are exactly $\langle a|a\rangle = (\Delta A)^2$ and $\langle b|b\rangle = (\Delta B)^2$, gives

$$|\langle\Delta\hat{A}\psi|\Delta\hat{B}\psi\rangle|^2 \leq (\Delta A)^2 (\Delta B)^2.$$

Since $\Delta\hat{A}$ is Hermitian, $\langle\Delta\hat{A}\psi|\Delta\hat{B}\psi\rangle = \langle\psi|\Delta\hat{A}\Delta\hat{B}|\psi\rangle = \langle\Delta\hat{A}\Delta\hat{B}\rangle$, so it remains to bound $|\langle\Delta\hat{A}\Delta\hat{B}\rangle|$ from below. Split the product into its symmetric and antisymmetric parts,

$$\Delta\hat{A}\Delta\hat{B} = \frac{1}{2}\{\Delta\hat{A}, \Delta\hat{B}\} + \frac{1}{2}[\Delta\hat{A}, \Delta\hat{B}],$$

and note that these two pieces have opposite Hermiticity properties: the anticommutator is Hermitian,

$$\{\Delta\hat{A}, \Delta\hat{B}\}^\dagger = (\Delta\hat{A}\Delta\hat{B})^\dagger + (\Delta\hat{B}\Delta\hat{A})^\dagger = \Delta\hat{B}\Delta\hat{A} + \Delta\hat{A}\Delta\hat{B} = \{\Delta\hat{A}, \Delta\hat{B}\},$$

while the commutator is anti-Hermitian,

$$[\Delta\hat{A}, \Delta\hat{B}]^\dagger = (\Delta\hat{A}\Delta\hat{B})^\dagger - (\Delta\hat{B}\Delta\hat{A})^\dagger = \Delta\hat{B}\Delta\hat{A} - \Delta\hat{A}\Delta\hat{B} = -[\Delta\hat{A}, \Delta\hat{B}].$$

The expectation value of a Hermitian operator is always real (this is just $i = j$ of the Hermitian eigenvalue argument of section 3.5, applied in an arbitrary, not necessarily eigen-, state: $\langle\hat{H}\rangle^* = \langle\psi|\hat{H}^\dagger|\psi\rangle = \langle\hat{H}\rangle$ when $\hat{H}^\dagger = \hat{H}$), while the expectation value of an anti-Hermitian operator is always purely imaginary, by the same argument with a sign flip. Also, since a c -number commutes with everything, $[\Delta\hat{A}, \Delta\hat{B}] = [\hat{A}, \hat{B}]$ exactly. Writing

$$x \equiv \frac{1}{2}\langle\{\Delta\hat{A}, \Delta\hat{B}\}\rangle \in \mathbb{R}, \quad \langle[\hat{A}, \hat{B}]\rangle \equiv 2iy, \quad y \in \mathbb{R},$$

we have $\langle \Delta \hat{A} \Delta \hat{B} \rangle = x + iy$, and therefore

$$|\langle \Delta \hat{A} \Delta \hat{B} \rangle|^2 = x^2 + y^2 \geq y^2 = \frac{1}{4} |\langle [\hat{A}, \hat{B}] \rangle|^2.$$

Combining this with the Cauchy–Schwarz bound above,

$$(\Delta A)^2 (\Delta B)^2 \geq |\langle \Delta \hat{A} \Delta \hat{B} \rangle|^2 \geq \frac{1}{4} |\langle [\hat{A}, \hat{B}] \rangle|^2,$$

and taking the (non-negative) square root of both sides gives the **generalized (Robertson) uncertainty principle**,

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|.$$

Two Hermitian operators can therefore be simultaneously assigned sharp (zero-uncertainty) values in the same state only if they commute in that state, and if $[\hat{A}, \hat{B}] \neq 0$ identically, the product of their uncertainties is bounded away from zero in *every* state. Specializing to position and momentum, whose commutator is fixed once and for all by the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar \hat{\mathbb{I}}$, gives the familiar

$$\Delta x \Delta p \geq \frac{\hbar}{2},$$

recovering the Heisenberg uncertainty relation as the special case $\hat{A} = \hat{x}$, $\hat{B} = \hat{p}$ of the general result derived above.

6.4 Simultaneous Diagonalizability and Complete Sets of Commuting Observables

section 4.13 already recorded the fact that commuting operators are simultaneously diagonalizable; we now prove this in full, including the degenerate case, together with its converse, and use it to make precise the notion — already anticipated by the appearance of a second label b_i within a degenerate eigenspace above — of exactly how many commuting observables are needed to specify a quantum state completely.

Non-degenerate case. Let $\hat{O}_1$ be Hermitian with a *non-degenerate* spectrum, eigenvectors $|a_i\rangle$ satisfying $\hat{O}_1 |a_i\rangle = a_i |a_i\rangle$ with all a_i distinct, and suppose $[\hat{O}_1, \hat{O}_2] = 0$ for some other operator $\hat{O}_2$. Then

$$\hat{O}_1 (\hat{O}_2 |a_i\rangle) = \hat{O}_2 \hat{O}_1 |a_i\rangle = a_i (\hat{O}_2 |a_i\rangle),$$

so $\hat{O}_2 |a_i\rangle$ is itself an eigenvector of $\hat{O}_1$ with eigenvalue a_i . Since a_i is non-degenerate, the corresponding eigenspace is one-dimensional and spanned by $|a_i\rangle$ alone, so $\hat{O}_2 |a_i\rangle$ must be proportional to $|a_i\rangle$,

$$\hat{O}_2 |a_i\rangle = b_i |a_i\rangle$$

for some scalar b_i : every eigenvector of $\hat{O}_1$ is automatically also an eigenvector of $\hat{O}_2$. As the $|a_i\rangle$ already form a complete orthonormal basis (by the Hermitian eigenvalue problem of section 3.5), this single basis diagonalizes $\hat{O}_1$ and $\hat{O}_2$ simultaneously.

Degenerate case. Suppose instead that $\hat{O}_1$ has a degenerate eigenvalue a , with eigenspace V_a of dimension $g > 1$ spanned by some orthonormal set $\{|a, 1\rangle, \dots, |a, g\rangle\}$, not assumed yet to be eigenvectors of $\hat{O}_2$. For any $|\phi\rangle \in V_a$,

$$\hat{O}_1 (\hat{O}_2 |\phi\rangle) = \hat{O}_2 \hat{O}_1 |\phi\rangle = a (\hat{O}_2 |\phi\rangle),$$

so $\hat{O}_2 |\phi\rangle \in V_a$ as well: $\hat{O}_2$ maps the eigenspace V_a into itself, i.e. V_a is an *invariant subspace* of $\hat{O}_2$. The restriction of $\hat{O}_2$ to V_a is itself a Hermitian operator on the finite-dimensional space V_a (Hermiticity of the restriction follows directly from Hermiticity of $\hat{O}_2$ on the whole space, since the inner product used to check it is the same one), and so, by the Hermitian

eigenvalue problem applied within V_a , it can be diagonalized there: V_a admits an orthonormal basis $\{|a, b_1\rangle, \dots, |a, b_g\rangle\}$ of joint eigenvectors,

$$\hat{O}_1 |a, b_k\rangle = a |a, b_k\rangle, \quad \hat{O}_2 |a, b_k\rangle = b_k |a, b_k\rangle, \quad k = 1, \dots, g.$$

Carrying out this diagonalization within *every* eigenspace of $\hat{O}_1$ and collecting the resulting bases produces an orthonormal basis of the whole space consisting entirely of joint eigenvectors of $\hat{O}_1$ and $\hat{O}_2$: the two operators are simultaneously diagonalizable exactly as in the non-degenerate case, the only difference being that a second label b_k , running over the (generally different) eigenvalues of $\hat{O}_2$ within each degenerate eigenspace of $\hat{O}_1$, is now needed to pick out a single common eigenvector.

Converse. Suppose conversely that $\hat{O}_1, \hat{O}_2$ admit a common orthonormal eigenbasis $\{|n\rangle\}$, $\hat{O}_1 |n\rangle = a_n |n\rangle$, $\hat{O}_2 |n\rangle = b_n |n\rangle$. Then for every basis vector,

$$[\hat{O}_1, \hat{O}_2] |n\rangle = \hat{O}_1 b_n |n\rangle - \hat{O}_2 a_n |n\rangle = a_n b_n |n\rangle - b_n a_n |n\rangle = 0,$$

and since $\{|n\rangle\}$ spans the whole space and $[\hat{O}_1, \hat{O}_2]$ is linear, $[\hat{O}_1, \hat{O}_2] |\psi\rangle = 0$ for every $|\psi\rangle$ in the space, i.e. $[\hat{O}_1, \hat{O}_2] = 0$ as an operator identity. Together with the two cases above, this establishes the full equivalence: two Hermitian operators commute if and only if they are simultaneously diagonalizable.

This immediately raises the question of how many mutually commuting observables are needed before a joint eigenvalue uniquely fixes the eigenvector (up to an overall phase), with no degeneracy left over. A set of mutually commuting Hermitian operators $\{\hat{O}_1, \hat{O}_2, \dots, \hat{O}_k\}$ (so that $[\hat{O}_\ell, \hat{O}_m] = 0$ for every pair, and hence, by repeated application of the argument above, all k admit a common eigenbasis) is called a **complete set of commuting observables**, or **CSCO**, if specifying the full list of eigenvalues $(a^{(1)}, a^{(2)}, \dots, a^{(k)})$ determines the corresponding joint eigenvector completely (again, up to an overall phase). A single non-degenerate Hermitian operator is trivially a CSCO by itself; the degenerate example just discussed shows the typical way a CSCO is built up in practice — starting from $\hat{O}_1$ alone, whose spectrum is degenerate, one adjoins a second commuting observable $\hat{O}_2$ whose eigenvalue b_k resolves the remaining degeneracy within each eigenspace of $\hat{O}_1$, and continues adjoining commuting observables until no degeneracy remains. Physically, a CSCO is exactly the list of observables that a maximal set of compatible (simultaneously measurable) quantities must furnish in order to prepare and label a quantum state completely: this is the algebraic content behind labeling atomic states by several quantum numbers at once, one for each operator in the relevant CSCO, rather than by a single number.

6.5 Normal Operators and the Spectral Theorem

section 3.5 and section 4.13 showed that a Hermitian operator always possesses a complete orthonormal eigenbasis, with real eigenvalues. It is worth asking exactly how much of this survives if Hermiticity, $\hat{O}^\dagger = \hat{O}$, is weakened to normality, $[\hat{O}, \hat{O}^\dagger] = 0$, the condition already introduced in section 3.5's neighborhood and noted there to hold for both Hermitian and unitary operators alike.

The key technical fact is a norm identity: for any normal operator $\hat{N}$ and any state $|\psi\rangle$,

$$\|\hat{N} |\psi\rangle\|^2 = \langle \psi | \hat{N}^\dagger \hat{N} | \psi \rangle = \langle \psi | \hat{N} \hat{N}^\dagger | \psi \rangle = \|\hat{N}^\dagger |\psi\rangle\|^2,$$

using normality, $\hat{N}^\dagger \hat{N} = \hat{N} \hat{N}^\dagger$, in the middle step: a normal operator and its adjoint annihilate exactly the same vectors, and more generally act on every vector with the same norm. Now let $\hat{O}$ be normal with $\hat{O} |a\rangle = \lambda |a\rangle$, and set $\hat{N} \equiv \hat{O} - \lambda \hat{\mathbb{I}}$. Since $\hat{O}$ is normal and a scalar multiple of the identity commutes with everything,

$$[\hat{N}, \hat{N}^\dagger] = [\hat{O} - \lambda \hat{\mathbb{I}}, \hat{O}^\dagger - \lambda^* \hat{\mathbb{I}}] = [\hat{O}, \hat{O}^\dagger] = 0,$$

so $\hat{N}$ is normal too, and by hypothesis $\hat{N}|a\rangle = 0$. The norm identity above, applied to $\hat{N}$ and $|a\rangle$, then gives $\|\hat{N}^\dagger|a\rangle\|^2 = \|\hat{N}|a\rangle\|^2 = 0$, so $\hat{N}^\dagger|a\rangle = 0$ as well, i.e.

$$(\hat{O}^\dagger - \lambda^*\hat{\mathbb{I}})|a\rangle = 0 \implies \hat{O}^\dagger|a\rangle = \lambda^*|a\rangle :$$

every eigenvector of a normal operator, with eigenvalue λ , is automatically an eigenvector of its adjoint, with eigenvalue λ^* — exactly the relation between $\hat{O}$ and $\hat{O}^\dagger$ that was simply given, for the special case $\hat{O}^\dagger = \hat{O}$, in the Hermitian eigenvalue problem of section 3.5.

This is already enough to redo the orthogonality argument of section 3.5 without assuming Hermiticity. Let $\hat{O}|a_i\rangle = \lambda_i|a_i\rangle$ and $\hat{O}|a_j\rangle = \lambda_j|a_j\rangle$ with $\lambda_i \neq \lambda_j$. Taking the adjoint of $\hat{O}^\dagger|a_j\rangle = \lambda_j^*|a_j\rangle$ gives $\langle a_j|\hat{O} = \lambda_j\langle a_j|$, and acting with this on $|a_i\rangle$,

$$\langle a_j|\hat{O}|a_i\rangle = \lambda_j\langle a_j|a_i\rangle.$$

But acting directly with $\hat{O}|a_i\rangle = \lambda_i|a_i\rangle$ gives instead $\langle a_j|\hat{O}|a_i\rangle = \lambda_i\langle a_j|a_i\rangle$. Subtracting,

$$(\lambda_i - \lambda_j)\langle a_j|a_i\rangle = 0 \implies \langle a_j|a_i\rangle = 0 \quad (\lambda_i \neq \lambda_j) :$$

eigenvectors of a normal operator belonging to distinct eigenvalues are orthogonal, exactly as for a Hermitian operator, even though the eigenvalues λ_i themselves need not be real. Within a degenerate eigenspace, Gram–Schmidt again produces an orthonormal set exactly as before. Collecting these facts: **a normal operator always possesses a complete orthonormal eigenbasis**, in which it takes the spectral form

$$\hat{O} = \sum_{i=1}^n \lambda_i |a_i\rangle\langle a_i|, \quad \lambda_i \in \mathbb{C},$$

of exactly the same shape as the Hermitian spectral decomposition of section 4.13’s neighborhood, but now with complex eigenvalues allowed. The converse holds too, and is immediate in this eigenbasis: if $\hat{O}$ has *some* complete orthonormal eigenbasis, then in that basis $\hat{O}^\dagger = \sum_i \lambda_i^* |a_i\rangle\langle a_i|$ is diagonal in the very same basis as $\hat{O}$, so $\hat{O}$ and $\hat{O}^\dagger$ certainly commute. This is the **spectral theorem**: an operator on a finite-dimensional inner-product space admits a complete orthonormal eigenbasis if and only if it is normal.

Normality, then, rather than Hermiticity, is the precise algebraic condition responsible for the existence of an orthonormal eigenbasis; Hermiticity is normality *plus* the extra requirement that the eigenvalues be real. Both unitary and Hermitian operators are normal, and both are unitarily diagonalizable, but only Hermitian operators are guaranteed real eigenvalues — a unitary operator’s eigenvalues lie on the unit circle, $\lambda = e^{i\theta}$, by the argument already given following the definition of normal matrices, and are real only in the special cases $\lambda = \pm 1$. Since a measured value of a physical observable must always come out as a real number, it is exactly this reality requirement, and not merely the desire for an orthonormal eigenbasis, that forces physical observables to be represented by Hermitian — rather than merely normal — operators.